

UNIVERSITY OF ZIMBABWE

SHORT-TERM MODELING AND ANALYSIS OF ZIMBABWE GOLD (ZIG) EXCHANGE RATES: AN EMPIRICAL STUDY USING TIME SERIES TECHNIQUES, VOLATILITY MODELING AND CHANGE POINT ANALYSIS

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To my caring mom,
my loving brother Tendai,
and everyone who has made it possible
for me to be here.

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List of Abbreviations

Abbreviation	Meaning
ARIMA	Auto-Regressive Integrated Moving Average
EGARCH	Exponential Generalized Autoregressive Conditional Heteroskedasticity
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
LSTM	Long Short-Term Memory
ML	Machine Learning
RBZ	Reserve Bank of Zimbabwe
RMSE	Root Mean Square Error
SVM	Support Vector Machine
USD	United States Dollar
ZiG	Zimbabwe Gold (currency)
ZWD	Zimbabwe Dollar
ZWL	Zimbabwe RTGS Dollar

Abstract

Abstract Information for "MODELLING AND FORECASTING THE DYNAMICS OF THE ZIMBABWE GOLD (ZIG)CURRENCY EXCHANGE RATES: AN EMPIRICAL ANALYSIS USING TIME SERIES AND STATISTICAL MODELING TECHNIQUES"

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Chapter 1

Introduction

1.1 Introduction

Exchange rates have a significant impact on trade balances, investment choices, and general financial stability, making them essential markers of a nation's economic health. Changes in these rates have a big impact on inflation and foreign investment. In order to manage the intricacies of global finance and preserve economic stability, economists and decision-makers must closely examine exchange rates.

An important turning point in Zimbabwe's economic history was the April 2024 introduction of the Zimbabwe Gold (ZiG) currency. In order to stabilize the nation's economy, which has traditionally experienced hyperinflation and currency volatility, the ZiG was introduced. The ZiG, which is supported by gold and foreign reserves, seeks to rebuild confidence in Zimbabwe's currency by providing a more reliable substitute for earlier monetary systems. Early assessments, however, indicate that the ZiG exchange rate has already seen significant fluctuations, beginning with a period of stability and ending with a strong increase in September 2024. This change points to underlying causes affecting the stability and volatility of the currency that should be investigated further.

The goal of this study is to fully examine the dynamics of the ZiG/USD exchange rate, with an emphasis on volatility analysis, change point detection, and short-term modeling. Using daily exchange rate data from the Reserve Bank of Zimbabwe (RBZ) since the introduction of the Zimbabwean dollar (ZiG), this study seeks to analyze the stability of the exchange rate, identify notable changes, and forecast short-term variations. In order to help policymakers make well-informed decisions about currency stability and economic resilience, the study looks at the levels and volatility of the ZiG exchange rate.

This study will fill in knowledge gaps on the ZiG's behavior, particularly as a recently established currency in a fragile economic climate. It will also emphasize how crucial it is to identify changes in exchange rate dynamics in order to make timely policy adjustments. By doing this, this study adds important empirical information about the ZiG and how it interacts with other economic variables. The project framework will be outlined in this chapter's conclusion, along with how next chapters will fulfill the study goals and support broader efforts to promote economic stability in Zimbabwe.

1.2 Research Background

Significant changes have occurred throughout Zimbabwe's currency history, including times of hyperinflation and continuous attempts to stabilize the economy. Introduced in 1970, the Rhodesian Dollar (RHD) was the currency of Southern Rhodesia until 1980, when the country gained independence and adopted the Zimbabwe Dollar (ZWD) (Global Financial Data, 2024). At first, the ZWD was more valuable than the USD, but over time, steep inflationary pressures reduced its value, leading to hyperinflation in 2007 and 2008, when prices doubled every day at their highest point (Chen, 2023). The demonetization of the ZWD in 2009 and the implementation of a multi-currency system, mostly based on the US dollar, were the results of this severe economic instability. While the multi-currency system offered stability, it constrained Zimbabwe's monetary policy flexibility and exposed the country to USD exchange rate volatility, which often misaligned with local economic demands (Makoni, 2019).

Zimbabwe tried to reestablish a sovereign currency in 2019 by combining bond notes and electronic money into the RTGS Dollar (ZWL). Despite being intended to bring about monetary independence, the ZWL was viewed with suspicion because of Zimbabwe's past currency problems. Low confidence and inflation continued to be major obstacles to stability, making attempts to float the ZWL unsuccessful (Muronzi, 2019). The economy consequently kept using the USD for day-to-day transactions, which reduced the ZWL's efficacy and indicated the country's reliance on foreign exchange (Nyathi & Dube, 2021).

In April 2024, the Zimbabwe Gold (ZiG) currency was introduced as the most recent attempt to fix Zimbabwe's monetary system. The ZiG is partially backed by gold reserves, in contrast to previous currencies, with the goal of fostering public confidence and establishing a stable, market-reflective monetary system within the framework of a flexible exchange rate. Despite supply and demand-driven fluctuations, the ZiG strives for stability with its gold backing (The Herald, 2024). At first, the ZiG/USD exchange rate hovered around 13 to 14 ZiG per USD. However, the official rate changed from 13.9 to 24.4 ZiG per USD on September 27, 2024, when the Reserve Bank of Zimbabwe depreciated the currency by 43%.

New exchange rate pressure brought on by rising food import prices and declining mineral export sales was blamed for this devaluation. The central bank chose to implement a currency adjustment instead of using reserves to preserve the prior rate in order to relieve this strain (Jusa, 2024). In light of shifting economic pressures, this incident emphasizes how crucial it is to investigate the underlying exchange rate dynamics of the ZiG.

Exchange rate analysis has emerged as a vital instrument for Zimbabwe's economic stability, as changes impact inflation, trade balances, and overall economic resilience. According to Yussif et al. (2021), exchange rate volatility can have a substantial impact on a country's import and export performance suggesting that changes may discourage foreign commerce or affect trading partners' pricing strategy. It is crucial to comprehend the ZiG's volatility in order to evaluate its impact on trade dynamics and economic stability.

This study's mainstay, time series analysis, offers a potent method for simulating short-term trends and patterns in currency swings. Clusters of volatility that represent market responses to economic situations can be captured when combined with volatility modeling, especially GARCH approaches. In line with the

objectives of this study, Yussif et al. (2021) point out that GARCH and EGARCH models are especially good at capturing abrupt changes in exchange rate behavior. For policymakers overseeing a new currency such as the ZiG, these insights can provide a projection of future times of stability or volatility, which is essential (Adams & Weiss, 2018; Lee & Kim, 2022).

Additionally, change point analysis is now useful for identifying changes in financial data series, which are frequently brought on by policy modifications or economic shocks. In their study, Manner, Rodríguez, and Stöckler (2024) demonstrate how change point analysis can detect notable changes in financial market volatility and dependence, underscoring its importance in comprehending the interdependencies of economic variables, especially in times of crisis. Change point identification in ZiG data may highlight important structural changes that are essential for currency stability initiatives, especially in light of Zimbabwe's unstable economic climate.

Although there has been some research on emerging market currencies, there is a noticeable lack of studies that are specifically to Zimbabwe, particularly with regard to its new currency, the ZiG. By employing daily exchange rate data to provide a thorough understanding of ZiG's short-term dynamics, this study seeks to close this gap. This study will uncover volatility patterns and possible changes in the stability of the ZiG by using time series analysis, volatility modeling, and change point analysis. This will provide important insights into Zimbabwe's policy framework. The ZiG's role in fostering economic recovery and strengthening monetary sovereignty in Zimbabwe is greatly supported by these findings.

1.3 Statement Of The Problem

An important attempt to stabilize an economy long plagued by hyperinflation and currency volatility was made with the introduction of the Zimbabwe Gold (ZiG) currency in April 2024. Recent changes in the exchange rate relative to the US dollar have sparked serious questions about the stability and efficacy of the ZiG as a monetary tool, despite the fact that it was expected to restore trust in the monetary system. A number of important issues can be identified:

1. **ZiG Exchange Rate Volatility:** The ZiG exchange rate has seen significant swings, climbing from between 13–14 ZiG per USD between April and August to as high as 24 ZiG per USD in September. The currency's intended function as a stable means of exchange is compromised by this volatility, which also puts investment and economic planning at risk (Chikohomero, 2024). Concerns regarding the currency's reliance on market conditions and the sufficiency of its gold backing to offset these swings are reflected in these oscillations.

2. **Early Shift Detection:** Policymakers must be able to spot early changes in the currency's behavior since these insights might yield important information for prompt remedial measures. Better economic decision-making can be supported by the capacity to identify exchange rate shift points, which can assist reduce potential hazards associated with abrupt fluctuations (Lustig, 2021).

3. **Economic Recovery Challenges:** The establishment of a stable and dependable currency is essential to Zimbabwe's continuous economic recovery efforts (Synergia Foundation, 2019). Attempts to draw in foreign investment and rebuild trust among local companies may be hampered by ignorance of the

ZiG's short-term movements and volatility patterns. These uncertainties could therefore make the nation's road to economic recovery even more difficult.

4. **Absence of Empirical Analysis:** The short-term dynamics of the ZiG have not been empirically studied, particularly in relation to time series analysis, volatility modeling, and the use of machine learning models for forecasting exchange rates. Using machine learning techniques can offer a fresh way to find intricate, non-linear patterns in the behavior of exchange rates, which could enrich or supplement knowledge gained from more conventional time series methods. There is a substantial knowledge gap on the ZiG's reaction to market forces and economic events as a result of the lack of studies that explicitly address the particular difficulties. Policymakers are unable to execute successful currency stabilization plans in the absence of empirical facts.

The purpose of this study is to thoroughly analyze the ZiG exchange rates utilizing time series methods, volatility modeling, and change point analysis in order to address these issues. Through recognizing trends in volatility and comprehending the elements impacting the ZiG's actions, this study will add significant understanding to Zimbabwe's economic structure and bolster the currency's function in fostering economic recuperation and augmenting monetary independence.

1.4 Research Questions

The study by the researcher intends to address the questions that follow:

1. What are the historical trends in the ZiG exchange rates against the USD since its introduction in April 2024?
2. Which time series modeling approaches are most capable of accurately predicting future changes in the ZiG/USD exchange rates?
3. What trends did the ZiG exchange rates exhibit in terms of volatility during the study period?
4. Can changes in the ZiG exchange rate's behavior be accurately detected using change point analysis, and what do these changes suggest about the stability of the currency?
5. What effects might the results have on Zimbabwe's economic strategy?

1.5 Research Objectives

This study's main objective is to use time series methods, volatility modeling, and change point analysis to examine the short-term dynamics of the Zimbabwe Gold (ZiG) exchange rates.

The particular goals are:

1. To examine the ZiG exchange rates' historical patterns in relation to the USD since their launch in April 2024.
2. To develop and apply both traditional time series models (e.g., ARIMA) and machine learning-based approaches (e.g., Long Short-Term Memory (LSTM) networks) to predict future changes in the ZiG/USD exchange rate, thereby assessing and comparing their effectiveness in capturing short-term exchange rate dynamics..

3. To look into the ZiG exchange rates' patterns of volatility during the study period.
4. To determine important changes in the ZiG exchange rate's behavior and evaluate their effects on currency stability using change point analysis.
5. To offer suggestions to decision-makers on the findings' significance for Zimbabwe's economic policy.

1.6 Research Hypothesis

This study will be guided by the following research hypotheses:

- **H1:** Since its launch in April 2024, the ZiG exchange rates versus the USD have shown notable historical tendencies.
- **H2:** Throughout the study period, the ZiG exchange rates show patterns of volatility that may be statistically examined and described.
- **H3:** Change point analysis is a useful tool for spotting notable changes in the ZiG exchange rate's behavior, which can point to unstable times.
- **H4:** Using both conventional time series modeling methods (like ARIMA) and machine learning-based strategies (like LSTM), future fluctuations in the ZiG/USD exchange rates can be accurately forecasted.
- **H5:** The study's conclusions will have a big impact on Zimbabwe's economic policies, especially when it comes to plans for economic recovery and currency stabilization.

1.7 Significance of the Study

Because the Zimbabwe Gold (ZiG) currency was only recently introduced, there hasn't been much prior research on its short-term dynamics, which this study thoroughly examines. This study provides numerous important contributions by concentrating on time series modeling, change point detection, and volatility patterns:

1. **Guidance for Financial Institutions and Investors:** This study offers crucial information for financial institutions, companies, and investors by illuminating ZiG's exchange rate trends and volatility. In an unstable market, having access to trustworthy data helps one make well-informed decisions (Nguyen, 2024).
2. **Improved Forecasting Strategies for Currency Management:**This paper presents sophisticated methods for forecasting currency behavior through the use of time series and machine learning techniques, including LSTM and ARIMA models. In order to promote proactive efforts to control possible volatility, these techniques seek to increase forecasting accuracy. Better short-term forecasts are supported by these models, which also aid in preventative actions to lessen possible volatility (Pay Set Limited, 2023).
3. **Policy Development and Currency Stabilization:** By helping policymakers pinpoint the causes of ZiG volatility, the study's conclusions will serve as a foundation for creating efficient monetary policies and interventions meant to stabilize the currency. This knowledge is essential for building investor trust and economic resilience.

4. **Emerging Market Economy Perspective:** Zimbabwe's currency shift is distinct from other emerging market economies. This study offers important insights into how to handle alternative and gold-backed currencies in comparable economic environments, and it might be used as a standard by other countries looking into similar approaches.

5. **Foundational Research on the ZiG:** This study creates a foundational body of information regarding the ZiG, as there is currently a dearth of empirical research on the subject. These results can be expanded upon in future studies on Zimbabwe's monetary policy and economic recovery, facilitating more investigation into the dynamics of currencies in unstable economies.

1.8 Organization of the Study

This study starts by presenting the research question concerning the Zimbabwe Gold (ZiG) currency and describing its importance in relation to short-term exchange rate dynamics, volatility modeling, and economic stability. The ZiG is placed within the larger framework of currency research in Chapter 2, which also offers a thorough literature assessment, highlighting any gaps that the study seeks to fill. Chapter 3 provides a full description of the methodology used, including sampling plans, data gathering methods, and analytic procedures, which support the methods used. In Chapter 4, the data analysis and results are presented. Time series analysis and change point identification are used to evaluate the volatility patterns of the ZiG.Chapter 5 ends with a summary of the results, their implications for investors, stakeholders, and policymakers, as well as suggestions for further study to further our understanding of how the ZiG has affected Zimbabwe's economic prospects.

1.9 Chapter Conclusion

The groundwork for the study of the Zimbabwe Gold (ZiG) currency has been established in this chapter, which also emphasizes the currency's importance in relation to Zimbabwe's economy and exchange rate forecasts. The study is to contribute to a better understanding of the behavior of the currency in a changing economic environment by analyzing the short-term dynamics of the ZiG with an emphasis on volatility patterns, time series modeling, and change point analysis. Highlighting the significance of this study, the chapter describes how the findings would benefit different stakeholders, such as investors, companies, and politicians. A thorough literature analysis will be provided in the following chapter, which will place this research within the body of existing scholarship and highlight any gaps that this study aims to fill. This will highlight the importance and urgency of investigating the dynamics of the ZiG.

Chapter 2

Literature Review

2.1 Introduction

This chapter examines the body of research to lay the groundwork for examining the short-term exchange rate dynamics of the Zimbabwe Gold (ZiG) currency, with an emphasis on volatility, change points, and pertinent statistical methods such as time series analysis and machine learning techniques. In light of the paucity of data on ZiG's performance since its launch in April 2024, this review places this study within larger exchange rate modeling discussions by examining theoretical frameworks and empirical investigations on comparable currency systems. The chapter ends with a conceptual framework that establishes the basis for an empirical investigation of the ZiG/USD exchange rate and connects the findings of the literature to the goals of the study.

2.2 Defination of Terms and Concepts

2.2.0.1 Exchange Rate

According to Hamilton (2018), an exchange rate is the price of one currency stated in terms of another currency or group of currencies. It is essential to international finance because it affects investment flows, trade balances, and economic stability (Zori, 2019). The USD-ZiG exchange rate is particularly important since the USD is widely utilized in Zimbabwean foreign exchange, affecting pricing, economic transactions, and financial stability, making it a vital tool for comprehending Zimbabwe's economic dynamics.

2.2.0.2 Time Series Analysis

"Time series analysis is a technique that enables analysts to see how variables change over time by examining a sequence of data points gathered at regular intervals" (Salesforce, Inc., 2022). In addition to revealing relationships between data points, this method offers insights into cyclical habits, trends, and seasonal patterns. For short-term currency rate modeling and forecasting, key models include ARIMA (Autoregressive Integrated Moving Average), Seasonal ARIMA (SARIMA), and Exponential Smoothing (ETS).By examining past data, these models create forecasts and give investors and policymakers important information about expected future changes in the ZiG/USD exchange rates.

2.2.0.3 ARIMA Model

ARIMA stands for Autoregressive Integrated Moving Average. According to Noble (2024), "it's a technique for time series analysis and for forecasting possible future values of a time series." ARIMA is used when a metric is recorded at regular intervals, ranging from fractions of a second to daily, weekly, or monthly periods (MastersInDataScience.org, 2023).

ARIMA models consist of the following components:

- **Autoregressive (AR)**: Denotes that an observation plus a predetermined number of lag observations are used in the model.
- **Integrated (I)**: Indicates how raw observations are differencing to make the time series stationary.
- **Moving Average (MA)**: This method uses a moving average model applied to lagged observations to model the relationship between an observation and residual errors.

The general formula for an ARIMA model, denoted as ARIMA(p, d, q), is:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q} + \epsilon_t \tag{2.1}$$

where:

- Y_t: The current value of the time series at time t.
- c: Constant term (if applicable).
- p: The order of the Autoregressive (AR) part, with parameters $\phi_1, \phi_2, \ldots, \phi_p$ that relate past values $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p}$ to the current value.
- d: The Integrated (I) component, indicating the number of differences needed to make the series stationary.
- q: The order of the Moving Average (MA) part, with parameters $\theta_1, \theta_2, \ldots, \theta_q$ that relate past errors $\epsilon_{t-1}, \epsilon_{t-2}, \ldots, \epsilon_{t-q}$ to the current value.
- ϵ_t: The error term (random shock) at time t.

Explanation of the Terms:

- **AR(p)**: The model includes p lagged values of Y_t, each weighted by ϕ coefficients.
- **I(d)**: The data is differenced d times to achieve stationarity.
- **MA(q)**: The model includes q lagged error terms, each weighted by θ coefficients.

2.2.0.4 SARIMA

SARIMA (Seasonal Auto-Regressive Integrated Moving Average) is an extension of the ARIMA (Autoregressive Integrated Moving Average) model that incorporates seasonality in addition to the non-seasonal components (Santra, 2023). It is denoted as ARIMA(p, d, q)(P, D, Q)[s] and the formula is:

$$Y_t = c+\phi_1 Y_{t-1}+\ldots+\phi_p Y_{t-p}+\theta_1\epsilon_{t-1}+\ldots+\theta_q\epsilon_{t-q}+\Phi_1 Y_{t-s}+\ldots+\Phi_P Y_{t-Ps}+\Theta_1\epsilon_{t-s}+\ldots+\Theta_Q\epsilon_{t-Qs}+\epsilon_t \quad (2.2)$$

where:

Non-seasonal components:

- p, d, q: Orders of the Autoregressive (AR), Integrated (I) (number of differences), and Moving Average (MA) terms, respectively.
- ϕ_i: Parameters of the non-seasonal AR terms.
- θ_j: Parameters of the non-seasonal MA terms.

Seasonal components:

- P, D, Q: Orders of the seasonal AR, differencing, and MA components.
- s: The seasonal period (e.g., 12 for monthly data with yearly seasonality).
- Φ_i: Parameters of the seasonal AR terms.
- Θ_j: Parameters of the seasonal MA terms.
- Y_t: The value of the time series at time t.
- ϵ_t: Error term (white noise) at time t.
- c: Constant term (if included).

2.2.0.5 Exponential Smoothing

Exponential Smoothing is a time series forecasting technique that predicts future values by assigning exponentially decreasing weights to past observations, giving more importance to recent data compared to older data (Malkari, 2023). The simplest form of an exponential smoothing formula is given by:

$$s_t = \alpha x_t + (1-\alpha)s_{t-1} = s_{t-1} + \alpha(x_t - s_{t-1}) \quad (2.3)$$

Here,

- s_t: Smoothed statistic; it is the simple weighted average of the current observation x_t.
- s_{t-1}: Previous smoothed statistic.
- α: Smoothing factor of data; where $0 < \alpha < 1$.
- t: Time period.

2.2.0.6 Machine Learning Time Series Model

Ruiz (2019) claims that these algorithms examine and forecast time-dependent data by utilizing machine learning approaches to identify patterns, trends, seasonality, and abnormalities. Since they can effectively manage complicated dependencies and non-linear interactions, models such as Random Forest, XGBoost, and LSTM networks are especially useful for short-term forecasting (Kurze, 2023).

H4: Using both conventional time series modeling methods (like ARIMA) and machine learning-based strategies (like LSTM), future fluctuations in the ZiG/USD exchange rates can be accurately forecasted.

2.2.0.7 Random Forest

Random Forest is a machine learning method that generates an ensemble of several decision trees in order to arrive at a single, more accurate prediction or outcome (Donges, 2024).

2.2.0.8 LSTM

LSTM (long Short Term Memory)is "an advanced form of Recurrent Neural Network, is crucial in Deep Learning for processing time series and sequential data"(Lawrence,2023). Because of this feature, LSTMs are especially useful for applications like forecasting, where it's crucial to identify patterns across time.

2.2.0.9 XGBOOST

A "scalable, distributed gradient-boosted decision tree (GBDT) machine learning library" (NVIDIA, 2023). XGBoost is well-known for its parallel tree boosting capabilities and is frequently used for forecasting jobs in addition to regression, classification, and ranking challenges. XGBoost is a useful tool for forecasting exchange rates and other financial time series variables because it efficiently models and forecasts time series data by utilizing its capacity to manage complicated dependencies and non-linear patterns(Wijaya,2023).

2.2.0.10 Volatility

Volatility is defined as "the degree of fluctuation in a currency's value over time, as indicated by the standard deviation of exchange rate returns" (Ilzetzki et al.,2023). The daily volatility formula is expressed as:

$$\text{Daily Volatility} = \sqrt{\text{Variance}} \tag{2.4}$$

where variance is calculated as:

$$\text{Variance} = \frac{1}{n}\sum(P_{\text{av}} - P_i)^2 \tag{2.5}$$

In this formula, P_i represents the daily exchange rate, P_{av} denotes the mean exchange rate, and n is the number of daily exchange rates.

To assess volatility on an annual basis, daily volatility is often scaled by the square root of 252, the approximate number of trading days in a year:

$$\text{Annualized Volatility} = \sqrt{252} \times \text{Daily Volatility} \tag{2.6}$$

Substituting the daily volatility formula, we have:

$$\text{Annualized Volatility} = \sqrt{252} \times \sqrt{\frac{1}{n}\sum (P_{\text{av}} - P_i)^2} \tag{2.7}$$

Exchange rate volatility is a crucial measure in evaluating financial risk, as it may indicate economic uncertainty or a currency's sensitivity to external shocks (Taşbaşı, Sarıca & Yüksel, 2020).

2.2.0.11 Volatility Modeling

Volatility modeling refers to the statistical techniques used to estimate and forecast fluctuations in exchange rates over time, enabling analysts to assess potential risks associated with currency movements (Epaphra, 2020). Models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) and EGARCH (Exponential GARCH) are particularly valuable in short term exchange rate analysis as they capture volatility clustering, providing crucial insights for investors and policymakers in assessing market risks and currency stability (Epaphra, 2020).

H2: Throughout the study period, the ZiG exchange rates show patterns of volatility that may be statistically examined and described.

2.2.0.12 GARCH Model

The GARCH model, which stands for Generalized Autoregressive Conditional Heteroskedasticity, "is a statistical model used to predict future volatility based on past data"(Ruppert, 2011). It estimates current volatility based on past periods' information, specifically focusing on the previous period's error terms and variances. The GARCH(1,1) model can be expressed with the following formulas:

Mean Equation:

$$r_t = \mu + \epsilon_t \tag{2.8}$$

Variance Equation:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \tag{2.9}$$

In these equations: - r_t represents the return at time t. - μ is the mean return. - ϵ_t is the error term (i.e., the shock). - σ_t^2 is the conditional variance at time t. - α_0, α_1, and β_1 are coefficients that need to be estimated.

2.2.0.13 EGARCH Model

The EGARCH model extends the GARCH framework to account for asymmetries in volatility, meaning it can capture situations where positive and negative shocks have different impacts on volatility. The EGARCH(1,1) model can be expressed as follows:

Mean Equation:

$$r_t = \mu + \epsilon_t \tag{2.10}$$

Logarithmic Variance Equation:

$$\log(\sigma_t^2) = \alpha_0 + \alpha_1 \left(\frac{\epsilon_{t-1}}{\sigma_{t-1}} \right) + \beta_1 \log(\sigma_{t-1}^2) \tag{2.11}$$

In these equations: - $\log(\sigma_t^2)$ is the logarithm of the conditional variance. - α_0, α_1, and β_1 are estimated parameters. - The term $\frac{\epsilon_{t-1}}{\sigma_{t-1}}$ allows for the modeling of leverage effects, where negative shocks can lead to higher future volatility compared to positive shocks of the same magnitude.

2.2.0.14 Change Point Analysis

It is a statistical technique for identifying notable changes in a time series, like abrupt variations in the mean, variance, or trend (Taylor, 2022). According to Manner et al. (2023), this method is useful for spotting structural changes in data that could point to significant policy or economic developments. Change point analysis, for example, might be used to assess if the September 2024 depreciation was a statistically significant shift in the ZiG/USD exchange rate, providing information about the stability of the currency and the efficacy of Zimbabwe's monetary policy.Its general framework;

1. **Define the Time Series:** Let Y_t be the time series of exchange rates at time t for $t = 1, 2, \ldots, T$.
2. **Model Parameters:** Assume there are k segments in the time series, defined by change points at $\tau_1, \tau_2, \ldots, \tau_{k-1}$, which partition the time series into segments.
3. **Likelihood Function:** The likelihood function can be defined as:

$$L(\theta) = \prod_{j=1}^{k} L_j(\theta_j) \tag{2.12}$$

   where $L_j(\theta_j)$ is the likelihood of segment j given its parameters θ_j.

4. **Change Point Detection:** To find the optimal change points, a common approach is to maximize the likelihood function or minimize the residual sum of squares (RSS) of the segments:

$$\text{RSS} = \sum_{j=1}^{k} \sum_{t \in S_j} (Y_t - \hat{Y}_{jt})^2 \tag{2.13}$$

   where S_j is the set of indices for segment j, and $\hat{Y}_{jt}$ is the predicted value for segment j.

5. **Statistical Test:** A statistical test, such as the Bayesian Change Point Detection or the CUSUM test, can be employed to evaluate the significance of detected change points.

H3: Change point analysis is a useful tool for spotting notable changes in the ZiG exchange rate's behavior, which can point to unstable times.

2.3 Theoretical Review

This section offers the theoretical foundations for examining the short-term dynamics of the Zimbabwe Gold (ZiG) currency's exchange rate, with an emphasis on complicated pattern detection, volatility modeling,

and structural change analysis. In order to effectively forecast the ZiG/USD exchange rate, these theories advocate the employment of GARCH and EGARCH models for volatility modelling, change point analysis, and machine learning techniques.

2.3.1 Theoretical Foundations of Time Series Analysis for Exchange Rate Forecasting

2.3.1.1 The Role of Historical Patterns in Forecasting

According to Hyndman and Athanasopoulos (2018), time series analysis is predicated on the idea that patterns found in historical data, such as trends, seasonal fluctuations, and cyclical behaviors, might provide important information about values in the future. Because historical patterns are frequently interpreted as representations of persistent market, political, and economic dynamics, this becomes particularly important when it comes to exchange rate forecasting (Tlegenova, 2015). Scholars and researchers can predict possible changes in future exchange rates by recognizing these recurrent patterns (Wei, 2006). The fundamental premise is that historical patterns would probably recur under generally stable circumstances, allowing analysts to use historical data as a solid foundation for forecasting both short-term and long-term changes in the future.

However, the fact that exchange rates are subject to unanticipated events, including policy changes or economic shocks, which might upset the established patterns, complicates this premise (Hamilton, 1994). Historical patterns can be useful for forecasting, but they should be used with caution, especially for volatile currencies where they might not be able to reliably anticipate future movements(Steinwart et al., 2012). In times of stability, time series models like ARIMA, SARIMA, and decomposition techniques have been crucial in spotting trends and cyclical tendencies. According to Box, Jenkins, and Reinsel (2008), these models aid researchers in determining whether transient variations reflect novel dynamics or are consistent with long-term patterns. In summary, historical trends can serve as a basis for forecasting, but their usefulness is reduced in settings characterized by instability or fast change.

2.3.1.2 Autoregressive Integrated Moving Average (ARIMA) Models

One of the most popular methods for univariate time series forecasting is the ARIMA model, which was first presented by Box and Jenkins in 1970. In stationary time series data, this approach works particularly well for spotting and forecasting linear trends and seasonal patterns. In order to convert the data into a stationary series, ARIMA combines three essential elements: differencing (I), moving average (MA) terms, and autoregressive (AR) terms (Reinsel et al., 2008). When dealing with non-stationary data, which is typical in exchange rate series, the differencing step is especially helpful. According to Ngan (2016), ARIMA is a useful technique for exchange rate prediction since it stabilizes volatility by lessening the influence of trends and seasonal changes, making it ideal for short-term forecasting.

2.3.1.3 Seasonal ARIMA (SARIMA) for Cyclical Patterns

Box and Jenkins (1976) introduced the Seasonal ARIMA (SARIMA) model, an extension of ARIMA that uses seasonal differencing to capture periodic changes in a time series, such those frequently seen in exchange rates. SARIMA is made to deal with recurrent patterns that appear on a yearly, quarterly, or monthly

basis and can be influenced by seasonal elements such policy changes, market activity, or economic cycles (Resende,2023). According to Etuk (2014), this approach is especially helpful in currency markets because exchange values are frequently influenced by seasonal cycles or other periodic changes. SARIMA is a useful method for assessing exchange rates that show seasonal variations since it can take these cyclical patterns into consideration.

2.3.1.4 State-Space Models and Kalman Filter for Adaptive Forecasting

Initiated as a theoretical formulation by Egon Beth (1961), state-space models were further developed by Bas van Fraassen (1970) and William Suppe (1988). These models offer a versatile and potent framework for forecasting, especially when data shows time-varying dynamics (WallStreetMojo, 2024). They are particularly helpful when market or regulatory changes result in sudden changes in exchange rate behaviors.State-space models depict the observable data as a function of unseen states that change over time, like underlying trends or abnormalities.

According to Durbin and Koopman (2012), the Kalman filter is a crucial part of these models. It is a recursive algorithm that estimates these unobserved states and updates them when fresh data becomes available. Since sudden fluctuations are common in short-term exchange rate forecasting, state-space models are quite useful due to their flexibility and adaptability, especially when combined with the Kalman filter.This paradigm is particularly useful in dynamic financial contexts because it can adapt to outside pressures that disrupt consistent patterns. According to Chukhrova (2021), the adaptive nature of state-space models enables more accurate forecasts even in the face of shifting circumstances in currency markets, where exchange values are susceptible to abrupt adjustments brought on by changes in the economy or in policy.

2.3.1.5 Decomposition Techniques for Trend and Seasonal Analysis

To break down a time series into its fundamental elements—trend, seasonality, and residuals—decomposition techniques are employed. According to Mahler (2023), by separating these components, decomposition techniques aid in comprehending the structure of exchange rate fluctuations and increase forecasting precision. Using locally weighted regression, STL (Seasonal-Trend decomposition using Loess) is one such technique that breaks down the series into trend, seasonal, and residual components (Cleveland et al., 1990). This method makes it easier to predict future movements by taking into account both short-term variations and long-term trends by assisting in the identification of seasonal and trend patterns within exchange rate data (Brownlee, 2020).

2.3.1.6 Forecasting Techniques for Single-Variable Time Series

Single-variable time series, like the ZiG/USD exchange rate, are especially well-suited for short-term forecasting. Methods like ARIMA, SARIMA, and decomposition techniques can be used to find past patterns and forecast future exchange rate changes. The flexibility of state-space models with Kalman filtering further improves forecasting accuracy by taking into account abrupt market movements and regulatory changes that can upset established patterns. These models are in line with the unpredictability of currency markets, which enables them to produce more accurate forecasts even when there is volatility.

2.3.2 Nonlinear Dynamics and Machine Learning Theory

Machine learning theory is based on the idea that systems may discover complex and frequently nonlinear patterns by learning from data to perform better on particular tasks (Roy & Rana, 2021). The concept of building artificial systems that replicate brain processing was sparked by McCulloch and Pitts' (1943) creation of the first mathematical model of neural networks. The Turing Test, which Alan Turing introduced as a standard in his groundbreaking book Computing Machinery and Intelligence, helped to further develop the idea of computer intelligence (Turing, 1950). The creation of the first self-learning program by Arthur Samuel in 1959 and the perceptron by Frank Rosenblatt, which represented a major advancement in the field of pattern recognition, came after these turning points (Rosenblatt, 1958).

Machine learning provides a strong foundation for capturing non-linear phenomena in financial time series analysis, especially for exchange rates. Conventional models like SARIMA and ARIMA work well for linear trends but have trouble with the intricacy of financial data, where economic shocks can cause abrupt changes and volatility clustering (Martin, 2019). Because they can represent long-term dependencies and non-linear interactions, machine learning models—especially those built on neural networks—perform exceptionally well in this field(Bonilla, 2023). Recurrent neural networks of the Long Short-Term Memory (LSTM) type are especially useful for time series forecasting because they can retain information over long sequences and are therefore skilled at identifying the complex patterns and temporal dependencies found in exchange rate data (Hochreiter & Schmidhuber, 1997).

Additionally, machine learning methods offer a foundation for adaptive modeling, which allows models to generalize from past data while taking into account both long-term trends and short-term volatility (Mohammadi, 2021). Given that fluctuations in the ZiG/USD exchange rate are characterized by volatility and possible non-linearity, this flexibility is particularly important for forecasting. In order to improve prediction accuracy and better handle the intricacies of financial markets, methods such as LSTM, Random Forest, and hybrid models may adapt dynamically to new data. Scholars can enhance the accuracy of exchange rate predictions by combining these approaches with traditional time series techniques, which capture the complex relationships that influence currency markets.

2.3.3 Volatility Clustering Theory

Mandelbrot (1963) first put forth the idea that "small changes tend to be followed by small changes, and large changes tend to be followed by large changes, of either sign." This concept is known as volatility clustering, an observed phenomenon in financial time series where periods of high volatility tend to be followed by high volatility, and low volatility by low volatility (Cont, 2005).The diagram below illustrates volatility clustering in time series data:

This persistent pattern reflects that volatility is serially correlated over time, influenced by external shocks, economic policies, and shifts in market sentiment (Cont, 2005). Behavioral factors, such as changes in investor analysis strategies, also contribute to this clustering, as investors respond to evolving market conditions (Suthar, 2023).

In exchange rate markets, volatility takes on various forms, influencing risk management and trading strategies. Key types include:

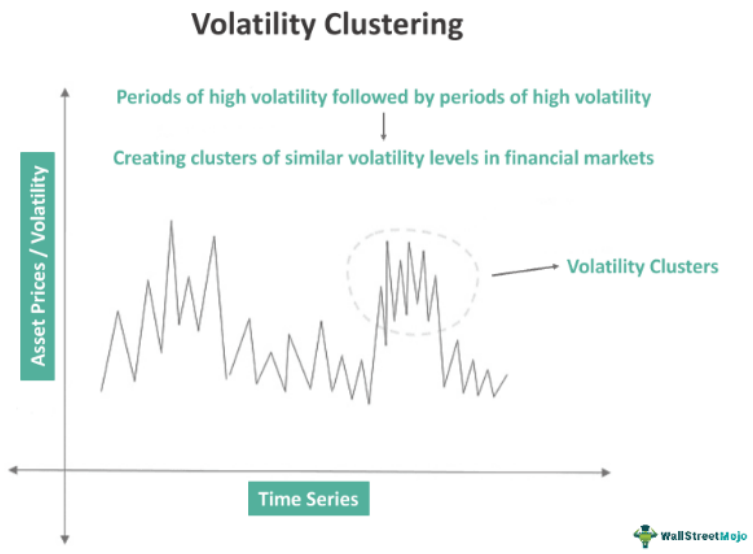


Figure 2.1: Illustration of Volatility Clustering in Time Series Data (Source: WallStreetMojo)

- **Historical Volatility:** Measures past fluctuations in exchange rates, providing a basis for predicting future movements.
- **Event-Driven Volatility:** Arises from unexpected events like economic reports or political announcements, often leading to sudden rate changes.
- **Currency Correlation Volatility:** Captures how exchange rate changes in one currency pair can impact others, reflecting the interconnected nature of global markets (FasterCapital, 2023).

These forms of volatility contribute to clustering as they create conditions where periods of high or low volatility persist, making it critical to use models that accurately capture this pattern. Models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity), developed by Engle (1982), allow conditional variances to evolve based on previous periods' volatility, adapting dynamically to market conditions. Bollerslev's (1986) EGARCH model extends this approach by modeling asymmetric responses to shocks, capturing situations where negative and positive events affect volatility differently. Such modeling flexibility is essential when analyzing exchange rate data like the ZiG/USD, where responses to economic events may be asymmetrical.

Nonlinear dynamics theories also acknowledge volatility clustering as a fundamental feature of financial markets that results from intricate relationships between market players and economic variables. This is explained by nonlinear dynamics in financial markets, where feedback mechanisms and information flow magnify market reactions (Xue-Zhong He, Kai Li, & Chuncheng Wang, 2016). This viewpoint encourages the use of possibly nonlinear models like LSTM for improved forecasting and sophisticated models like GARCH and EGARCH to capture volatility clustering.

Assessing currency stability in the context of the ZiG/USD exchange rate requires an understanding of volatility clustering. GARCH and EGARCH models are useful for determining times of high or low stability since economic uncertainties and external shocks can cause extremely volatile movements (Engle,

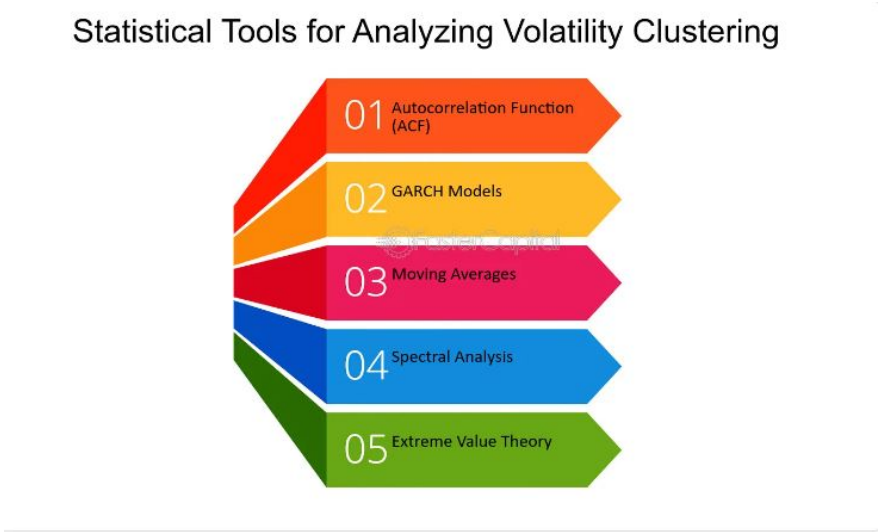


Figure 2.2: Statistical Tools for Analyzing Volatility Clustering (Source: FasterCapital)

1982; Bollerslev, 1986). By quantifying stability patterns, these models shed light on how resilient the ZiG is to changes in the economy.

For investors, policymakers, and analysts, understanding volatility clustering in exchange rates is crucial because it helps them spot possible hazards, optimize asset allocation, and direct their investment and risk management plans. Because of their vulnerability to global market dynamics, policy changes, and economic shocks, exchange rates are especially vulnerable to volatility clustering. Financial professionals can effectively manage currency risk, make well-informed decisions about asset allocation, and modify investment strategies in response to shifting market conditions by having a thorough grasp of the clustering of exchange rate volatility (Sahu, 2023).

2.3.4 Structural Break Theory

Andrews (1993) was the first to officially define the notion of structural breaks, offering statistical criteria for identifying sudden and significant changes in time series that alter their basic properties. These structural fractures, which are frequently brought on by outside shocks, changes in policy, or noteworthy economic events, have an effect on the forecasting models' accuracy, according to Hansen (2001). These changes bring about times when the historical behavior of the time series is no longer relevant. Because they signal regime changes that could impact asset pricing, volatility, and market risk, structural breaks are particularly important in financial markets (Perron, 2006).

In exchange rate modeling, it is crucial to recognize and take into consideration structural breaks since they have the potential to unexpectedly modify the dynamics of currency values. Currency values may undergo structural changes as a result of changes in government policy, such as monetary or exchange rate policy (Bai & Perron, 2003). Periods of increased volatility are frequently the consequence of such shifts, and conventional models that do not account for structural discontinuities may produce inaccurate projections (Chen & Patel, 1998). By quantitatively identifying breakpoints instead of depending on conjecture, techniques such as the Bai-Perron test offer a statistically rigorous approach to breakpoint detection, increasing

model robustness (Zarei et al., 2015).

2.3.4.1 Change Point Detection (CPD) in Structural Break Analysis

One important technique for locating structural fractures is change point detection (CPD). It entails testing two hypotheses: the alternative hypothesis (H_a), which contends that a change has taken place at a particular moment, and the null hypothesis (H_0), which holds that no change happens(Preis et al.,2023). The CPD procedure provides a statistical foundation for model adjustment by assisting in determining the precise point at which a time series' characteristics change (Chen & Zhang, 2015).

Change Point Detection Hypotheses:

- H_0: Throughout the interval (m, n), the time series does not change.
- H_a: The characteristics of the time series change at a change point (k^*).

Where P_{X_i} denotes the probability density function at point x_i, and k^* represents the change point. This paradigm helps detect abrupt regime shifts by serving as the foundation for statistical studies that find structural discontinuities in exchange rate data.A simple illustration of a change-point detection task can help visualize this process, as shown below:

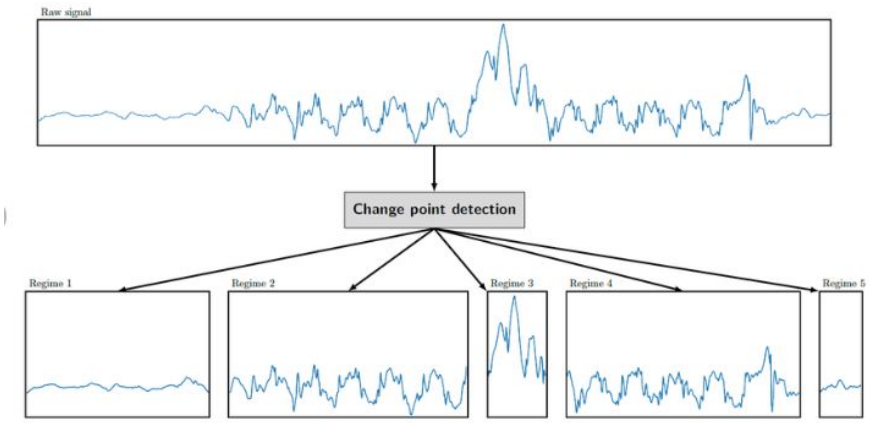


Figure 2.3: Simple illustration of a change-point detection task

2.3.4.2 Types of Change Points in Time Series and Their Relevance to Exchange Rate Forecasting

Identifying and comprehending the various kinds of change points in time series data is essential for exchange rate forecasting. Change points are times when a time series' statistical characteristics change significantly; by taking into account the changing nature of exchange rates, these shifts enable more accurate modeling and forecasting. Christof Schroth, Dr. Julien Siebert, and Janek Grob (2021) distinguish a number of different types of change points that can appear in a time series:

1. **Change Point in Mean:** A variation in the series' average value. This kind of change point could signal a significant economic event that affects the overall level of exchange rates, such as the launch of a new currency or a major policy change.

2. **Change Point in Variance:** A modification to the series' variability or dispersion. This change point could indicate rising market volatility, which may be triggered by abrupt geopolitical tensions, shifts in international trade regulations, or global financial crises.

3. **Change Point in Frequency:** A change in the data's periodicity or frequency. This may occur when the dynamics of the currency rate are influenced by external factors, such as shifts in trading hours, foreign market schedules, or seasonal influences.

4. **Change Point in Pattern:** A modification to the series' underlying pattern or trend. This often occurs when there is a long-term shift in the behavior of exchange rates due to changes in the economy, such as alterations in interest rates, inflation rates, or structural adjustments.

5. **Change Point in Multidimensional Time Series:** A collection of time series that interact in a multivariate context. This includes interactions between exchange rates and other economic factors, such as interest rates or commodity prices, or shifts that affect multiple currency pairs simultaneously.

A more sophisticated view of structural changes in exchange rate data is offered by these kinds of change points. Determining the precise kind of change that has taken place can help tailor the study to more accurately reflect its many facets, whether they are connected to market dynamics, economic events, or policy initiatives(Booras,2021).Since it enables the model to adjust to various time series regimes, identifying and simulating these shift points in exchange rate analysis can result in more accurate projections. For example, geopolitical events or economic shocks, including the implementation of new governmental policies or shifts in the prices of important commodities, may be accompanied by notable changes in exchange rates. Eklou (2023) asserts that government actions or reactions to modifications in the dynamics of imports and exports, as well as changes in the price of commodities globally, may cause structural breaks in exchange rates.Analysts can create more reliable models that faithfully capture the dynamic and frequently erratic nature of exchange rate fluctuations by comprehending and recognizing the various kinds of change points.

2.3.4.3 Practical Applications of Change Point Analysis in Exchange Rate Forecasting

For currencies like the Zimbabwean Gold (ZiG), change point analysis is a useful tool for improving exchange rate forecasts. Through the discovery of data shifts or breakpoints, this technique makes it possible to identify fundamental changes in the exchange rate, which forecasting models can then modify appropriately. Such fluctuations in the ZiG/USD exchange rate can be brought about by external forces or adjustments in economic policy, and identifying these breakpoints increases the accuracy of forecasts (Yussif et al., 2021).

Applying change point analysis to exchange rate forecasting enables the data to be divided into discrete regimes, each with unique properties. By ensuring that the forecast takes into account the currency rate's shifting dynamics over time, this segmentation increases model accuracy. Finding these structural changes improves the model's ability to estimate how the ZiG will react to macroeconomic events and offers more reliable predictions when the economy is changing.

2.4 Statistical Tests in Time Series Analysis and Forecasting

According to Brookwell (2023), statistical tests are essential to time series analysis since they verify important hypotheses, identify possible problems, and guarantee the accuracy of forecasting models. When using models like ARIMA or GARCH, these tests assist in determining whether the data satisfies basic characteristics such stationarity and the lack of autocorrelation (Topas, 2024). In order to help researchers choose the best modeling approaches, they also detect structural fractures, cointegration linkages, and volatility clustering (Yen, 2023). These tests are especially important for recently issued currencies, such as the Zimbabwe Gold (ZiG), when data is scarce and exchange rates are subject to instability.They improve the accuracy of exchange rate projections by offering a framework to deal with issues including data sparsity, non-stationarity, and abrupt regime adjustments.

2.4.1 Stationarity Tests

A time series is said to be stationary if its observations are independent of the time at which they are recorded (Krishnan, 2021). A stationary time series is stable and predictable for analysis since its statistical characteristics, such as mean, variance, and covariance, don't change over time.

2.4.1.1 Augmented Dickey-Fuller (ADF) Test

According to machinelearningplus (2019), the Augmented Dickey-Fuller (ADF) test is a popular statistical method used to assess whether a time series is stationary or has a unit root, indicating non-stationarity. In time series analysis, this test is essential since stationarity is a fundamental premise of many forecasting models. An effective paradigm for assessing stationarity is provided by the ADF test, which uses lagged differences to account for higher-order autocorrelations.Formula;

$$\text{ADF Test} = \Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta Y_{t-i} + \epsilon_t \tag{2.14}$$

Where:

- Y_t: The time series being tested.
- ΔY_t: The first difference of Y_t ($Y_t - Y_{t-1}$).
- α: Intercept term.
- βt: Trend term (optional, included if testing for trend stationarity).
- γY_{t-1}: Lagged value of the series.
- $\sum_{i=1}^{p} \delta_i \Delta Y_{t-i}$: Lagged differences to account for higher-order autocorrelation.
- ϵ_t: Error term.

The null hypothesis (H_0) of the ADF test states that the time series has a unit root (i.e., it is non-stationary). If the test statistic is less than the critical value, the null hypothesis is rejected, indicating stationarity.

2.4.1.2 KPSS Test

Kwiatkowski et al. (1992) developed the KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test, a statistical method for determining if a given time series is stationary around a deterministic trend. The KPSS test accepts stationarity as the null hypothesis, in contrast to other unit root tests, such as the ADF, which test for non-stationarity as the null hypothesis. Because of this, it provides an additional tool for confirming the stationarity of a time series and is especially helpful in figuring out whether de-trending or differencing is required.The following breakdown of a time series serves as the foundation for the KPSS test:

$$Y_t = r_t + \beta t + \epsilon_t \tag{2.15}$$

Where:

- Y_t: Observed time series.
- r_t: Stationary process (the test evaluates this component).
- βt: Deterministic trend.
- ϵ_t: Random error term with a mean of zero and constant variance.

The test statistic is calculated as:

$$\text{KPSS} = \frac{1}{T^2 \hat{\sigma}^2} \sum_{t=1}^{T} S_t^2 \tag{2.16}$$

Where:

- S_t: Cumulative sum of residuals, defined as $S_t = \sum_{i=1}^{t} \hat{\epsilon}_i$.
- $\hat{\sigma}^2$: Long-run variance of the residuals.
- T: Number of observations in the series.

Interpretation:

- Null Hypothesis (H_0): The series is stationary.
- Alternative Hypothesis (H_1): The series is non-stationary.

2.4.1.3 Phillips-Perron (PP) Test

Developed by Phillips and Perron (1988), the Phillips-Perron (PP) test is a unit root test used to determine if a time series is non-stationary. It can be used as a substitute for the Augmented Dickey-Fuller (ADF) test, but it takes a different approach to accounting for heteroskedasticity and serial correlation. As opposed to the ADF test, the PP test employs a nonparametric adjustment to the test statistic and does not call for the addition of lagged difference terms.The PP test is especially helpful when dealing with time series data that can contain time-varying variances or autocorrelated errors.The PP test is based on the following autoregressive model:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \epsilon_t \tag{2.17}$$

Where:

- Y_t: Time series being tested.
- ΔY_t: First difference of the time series ($Y_t - Y_{t-1}$).
- α: Constant (drift term).
- β: Coefficient on time t (trend component).
- γ: Coefficient tested to check for a unit root ($H_0 : \gamma = 0$).
- ϵ_t: Error term.

The test statistic is calculated as:

$$t_\gamma = \frac{\hat{\gamma}}{SE(\hat{\gamma})} \tag{2.18}$$

Where $SE(\hat{\gamma})$ is the standard error of the estimated coefficient γ.
Interpretation:

- Null Hypothesis (H_0): The series has a unit root (non-stationary).
- Alternative Hypothesis (H_1): The series does not have a unit root (stationary).

If the test statistic is more negative than the critical value, the null hypothesis is rejected, suggesting stationarity.

2.4.2 Cointegration Tests

According to Engle and Granger (1987), cointegration is the process by which non-stationary variables are linearly joined to create a stationary connection that implies a long-term equilibrium between them. In time series analysis, this is essential because it enables researchers to simulate the dynamics of variables that, in spite of individual trends, have a constant relationship over time. Understanding equilibrium adjustments is crucial for precise forecasting and policy formation, and identifying cointegration aids in this process.

2.4.2.1 Engle-Granger Test

Engle and Granger (1987) created the two-step Engle-Granger test, which is used to determine if two or more non-stationary time series are cointegrated. In order to test the residuals for stationarity, a unit root test, like the Augmented Dickey-Fuller (ADF) test, is used after estimating a linear relationship between the variables using ordinary least squares (OLS). There may be a cointegrating relationship between the variables, indicating that they share a long-term equilibrium, if the residuals are stable.Particularly when there are just two variables, this is the most straightforward and popular cointegration test. There are two steps involved:

1. **Regression:** The two variables are first regressed against one another.
2. **Residual Testing:** A unit root test, such as the Augmented Dickey-Fuller (ADF) test, is then used to check for stationarity in the regression's residuals. The two variables are assumed to be cointegrated if the residuals are stationary.

2.4.3 Model Diagnostics

Model diagnostics are vital for making sure the forecasting model selected fits the data correctly and satisfies fundamental presumptions (Rudolph,2023). For confirming key presumptions, including independence, normalcy, and homoscedasticity, residual analysis is very crucial. It is preferable for residuals—which show the variations between the observed and expected values—to be dispersed randomly and devoid of any discernible trends. It may be necessary to improve the model further if the residuals show patterns, which indicate that the model is not fully capturing the data. Assessing the model's suitability, spotting possible problems like misspecification, and enhancing its forecasting capabilities are all made possible by appropriate diagnostic checks (Koehler & Murphree, 2018).

2.4.3.1 Ljung-Box Q-Test

The autocorrelation of residuals over various time lags is evaluated using the Ljung-Box Q-Test (Bobbitt, 2020). It examines the independent distribution of residuals, which is the null hypothesis. Significant autocorrelation suggests that the model has not been able to identify a pattern in the data. In time series analysis, where autocorrelation is prevalent and can affect the model's predicting accuracy, this test is very helpful (Ljung & Box, 1978).It is based on the following formula:

$$Q = n(n+2)\sum_{k=1}^{m}\hat{\rho}_k^2 \tag{2.19}$$

Where:

- n: Sample size (number of observations).
- m: Number of lags being tested.
- $\hat{\rho}_k$: Sample autocorrelation at lag k.

The sum runs over the number of lags m.

The test statistic Q follows a chi-squared distribution with m degrees of freedom under the null hypothesis of no autocorrelation.

2.4.3.2 Durbin-Watson Statistic

Regression model residuals can be examined for serial correlation using the Durbin-Watson statistic (Prep-Nuggets, 2023). Significant values below or above 2 indicate positive or negative serial correlation, respectively, whereas a value close to 2 indicates no autocorrelation. Unreliable forecast intervals and skewed parameter estimates can result from serial correlation in residuals (Durbin & Watson, 1951). When determining if model mistakes are randomly distributed, this test is crucial in time series forecasting.The formula for the Durbin-Watson statistic is;

$$DW = \frac{\sum_{t=1}^{n} e_t^2}{\sum_{t=2}^{n}(e_t - e_{t-1})^2} \tag{2.20}$$

Where:

- e_t: Residual at time t.

- e_{t-1}: Residual at time $t-1$.
- n: Number of observations.

Interpretation:

- Null Hypothesis: There is no first-order autocorrelation in the residuals (i.e., the residuals are independent).
- Alternative Hypothesis: There is significant first-order autocorrelation in the residuals.

2.4.3.3 ARCH-LM Test

In order to determine whether the variance of residuals depends on past error terms, the Autoregressive Conditional Heteroscedasticity (ARCH) LM test looks for conditional heteroscedasticity, a feature frequently observed in financial time series data (Yasar, 2021). If significant heteroscedasticity is present, it suggests that the model does not adequately account for varying volatility over time, which is particularly important in exchange rate forecasting, where volatility clustering (high volatility followed by low volatility) is a common phenomenon (Engle, 1982).The general form of the test involves the following steps:

1. Estimate the regression model (e.g., AR model) and compute the residuals e_t.
2. Square the residuals e_t^2, and regress these squared residuals on a set of lagged squared residuals:

$$e_t^2 = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 e_{t-2}^2 + \cdots + \alpha_q e_{t-q}^2 + \epsilon_t \tag{2.21}$$

   where q is the number of lags in the ARCH model (usually denoted as the order of the ARCH effect).

3. Obtain the Lagrange Multiplier (LM) statistic, which is based on the residual sum of squares from the regression:

$$LM = T \cdot R^2 \tag{2.22}$$

   where:

   - T: Number of observations (sample size).
   - R^2: Coefficient of determination from the regression of squared residuals on the lagged squared residuals.

Hypothesis Testing:

- Null Hypothesis (H_0): No ARCH effect (no conditional heteroscedasticity).
- Alternative Hypothesis (H_1): The presence of ARCH effect (conditional heteroscedasticity exists).

2.5 Statistical Tests in Volatility Modeling

Key traits that are essential to models like GARCH, such as heteroscedasticity or volatility clustering, are verified by statistical tests in volatility modeling. These tests guarantee that the model of choice incorporates the natural grouping and variability found in financial data. Volatility modeling statistical tests are described in this section.

2.5.0.1 Engle's ARCH Test

The ARCH test, developed by Engle in 1982, is essential for determining whether autoregressive conditional heteroscedasticity (ARCH) effects are present in time series data. The variance of the errors is assumed to be dependent on previous error terms rather than being constant in ARCH models.The formula is;

$$\epsilon_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + u_t \tag{2.23}$$

Where:

- ϵ_t^2: Squared residuals from the model.
- α_0: Intercept.
- α_i: Coefficients of lagged squared residuals.
- u_t: Error term.

The null hypothesis (H_0) is that $\alpha_i = 0$, indicating no ARCH effect.

When huge or little price swings are followed by similarly large or small movements, this test is essential for determining whether volatility clustering is present. When it comes to exchange rates such as the ZiG/USD, volatility clustering is frequently seen during times of market stress or noteworthy economic developments.Researchers can verify whether exchange rate data volatility is time-varying by using Engle's ARCH test. They can also determine whether more complex models of volatility, such as Generalized Autoregressive Conditional Heteroscedasticity (GARCH), should be employed for precise volatility forecasting and risk management (Engle, 1982).

2.5.0.2 Box-Ljung Test for Residuals

The Ljung-Box test for residuals is a statistical test developed by Greta M. Ljung and George E. P. Box in 1978 to ascertain whether a time series' autocorrelations deviate considerably from zero. This test is frequently used to determine whether autocorrelation exists in residuals following the fitting of a time series model, as Hyndman (2023) points out. Here is the test statistic:

$$Q^* = T(T+2) \sum_{k=h}^{h} (T-k)^{-1} r_k^2 \tag{2.24}$$

Where:

- T: Length of the time series.
- r_k: The kth autocorrelation coefficient of the residuals.
- h: Number of lags to test.

A large value of Q^* indicates that there are significant autocorrelations in the residual series.

2.5.0.3 Likelihood Ratio Test for GARCH Models

The Likelihood Ratio (LR) test determines if a more intricate GARCH model fits data better than a more straightforward ARCH model (Burch, 2005). It tests the null hypothesis that the simpler model is adequate by comparing the probabilities of the two models. This is the test statistic:

$$LR = 2\left[\ln(L_{\text{GARCH}}) - \ln(L_{\text{ARCH}})\right] \tag{2.25}$$

Where:

- $\ln(L_{\text{GARCH}})$: Log-likelihood of the GARCH model.
- $\ln(L_{\text{ARCH}})$: Log-likelihood of the ARCH model.

This test compares the goodness of fit between the GARCH and ARCH models, with a higher likelihood indicating a better fit.With degrees of freedom equal to the variation in the number of parameters, the statistic is distributed according to a chi-square model. This test is essential for forecasting exchange rates since it makes sure the chosen model accurately depicts volatility dynamics, increasing prediction accuracy.

2.5.0.4 Jarque-Bera Test for Normality

A statistical test called the Jarque-Bera (JB) test is used to determine if the regression model's residuals are normally distributed (Thadewald, 2004). Many econometric models make the assumption that residuals are normal. This test determines whether the data's skewness (asymmetry) and kurtosis (tailedness) deviate from a normal distribution (Aslam, 2021). In time series analysis, particularly exchange rate modeling, the JB test is frequently used because non-normal residuals may be a sign of model misspecification. It indicates that model selection has to be changed or that more robust models that do not rely on normalcy assumptions should be used if residuals in exchange rate forecasting models are not normally distributed. Here's how to calculate the JB test statistic:

$$JB = \frac{6}{n}\left(S^2 + 4(K-3)^2\right) \tag{2.26}$$

Where:

- n: Number of observations.
- S: Skewness of the residuals.
- K: Kurtosis of the residuals.

The null hypothesis for the JB test is that the residuals are normally distributed. If the test statistic exceeds the critical value from the chi-square distribution, the null hypothesis is rejected, indicating that the residuals are not normally distributed.

2.5.0.5 Relevance to Exchange Rates

To find out if the data satisfies the assumptions needed to use ARIMA models, stationarity tests like the Augmented Dickey-Fuller (ADF) and KPSS are crucial. These tests determine if the series' mean and variance remain constant over time; the KPSS test confirms stationarity, while the ADF test looks for the

presence of a unit root. Before moving on to ARIMA modeling, differencing will be used to bring non-stationary data into stationarity. These tests provide a strong basis for precise short-term exchange rate forecasts by meeting the stationarity criterion.

Furthermore, tests such as the ARCH-LM and Bai-Perron offer more information about the particular dynamics of the ZiG/USD exchange rate. The use of GARCH models to capture time-varying volatility patterns is justified by the ARCH-LM test, which detects volatility clustering. In the meanwhile, major regime shifts brought on by modifications to policies or outside shocks are identified using structural break tests like the Chow and Bai-Perron tests. By dividing the data into discrete stable and unstable periods, these tools make sure that the models chosen accurately represent the economic conditions of each regime. When combined, these statistical tests guarantee that the methodology takes into consideration the ZiG's changing market conditions, resulting in models that are accurate and flexible.

2.6 Statistical Tests in Change Point Analysis

Change point analysis, as explained by Li (2021), is essential for identifying times of instability that may affect forecasting accuracy and for recognizing structural discontinuities in exchange rate behavior. These statistical analyses assess if there are notable changes in the time series' mean, variance, or other characteristics at particular intervals. For the development of models that adjust to changes in market dynamics, such insights are essential. In order to forecast exchange rates, change point analysis uses statistical tests, which are explained in this section.

2.6.0.1 Chow Test

Gregory Chow (1960) developed the Chow Test, a statistical method for identifying if a structural break exists at a specific moment in time. This test contrasts two distinct regression models: one that uses all of the data and another that divides the data into two subsets: the presumptive break and the post-break data.Its equation:

$$F = \frac{(RSS_{\text{pooled}} - (RSS_1 + RSS_2))/k}{(RSS_1 + RSS_2)/(n_1 + n_2 - 2k)} \tag{2.27}$$

Where:

- RSS_{pooled}: Residual sum of squares from the pooled model (using the entire dataset).
- RSS_1: Residual sum of squares from the regression model for the first subset of the data.
- RSS_2: Residual sum of squares from the regression model for the second subset of the data.
- k: Number of parameters in the model.
- n_1: Number of observations in the first subset.
- n_2: Number of observations in the second subset.

The Chow test essentially checks if the coefficients of the two models are significantly different from each other. A significant F-statistic indicates the presence of a structural break.

Relevance for Exchange Rates: In exchange rate analysis, the Chow Test can help identify when shifts in economic policies or external shocks lead to significant breaks in the behavior of exchange rates.

2.6.0.2 Bai-Perron Test

Bai and Perron (1998) introduced the Bai-Perron Test, which is used to identify multiple structural breakdowns at unknown times. The Bai-Perron test enables the identification of many change points in the data, in contrast to the Chow Test, which presumes a single break at a specified point.By minimizing the sum of squared residuals across various break points, the Bai-Perron method finds the break points that minimize this value. In terms of mathematics, it can be stated as:

$$\hat{\theta} = \arg\min \left(\sum_{i=1}^{m+1} \sum_{t=\theta_i}^{\theta_{i+1}} \epsilon_t^2 \right) \tag{2.28}$$

Where:

- $\hat{\theta}$: Estimated break points.
- θ_i: Time points where breaks occur.
- ϵ_t: Residual for time point t.

By segmenting the time series according to the identified break points, this technique models each fragment independently.

Relevance to Exchange Rates: When there are several changes in the political or economic landscape over time, the Bai-Perron Test can be especially helpful in predicting exchange rates. This test offers a more thorough comprehension of the potential behavior of exchange rates over time.

2.6.0.3 Cumulative Sum Control Chart (CUSUM) Test

Page (1954) developed the CUSUM test, which tracks cumulative sums of deviations from the mean to identify changes in a time series' mean level. When the cumulative total of a regression model's residuals beyond specific critical thresholds, the test detects a possible structural break.Its equation is:

$$S_t = \sum_{i=1}^{t} (\epsilon_i - \bar{\epsilon}) \tag{2.29}$$

Where:

- S_t: Cumulative sum at time t.
- ϵ_i: Residual at time i.
- $\bar{\epsilon}$: Mean residual.

If S_t exceeds a predefined threshold (either positive or negative), a change point is detected. **Relevance to Exchange Rates:**The CUSUM test is particularly helpful for tracking changes in exchange rates that could be brought on by modifications to policy, outside shocks, or other systemic issues that impact the behavior of the currency.

2.6.0.4 Piecewise Linear Regression (Segmented Regression)

Piecewise or segmented linear regression is the process of fitting distinct linear regression models to various time series parts. When the regression line's intercept or slope varies between two segments, it is said to have reached a change point.The following is the equation for a piecewise linear regression model:

$$y_t = \begin{cases} \beta_0 + \beta_1 t + \epsilon_t & \text{for } t < \tau \\ \gamma_0 + \gamma_1 t + \epsilon_t & \text{for } t \geq \tau \end{cases} \tag{2.30}$$

Where:

- β_0, β_1: Parameters of the linear model before the change point τ.
- γ_0, γ_1: Parameters of the linear model after the change point τ.
- ϵ_t: Residual error term.

The objective is to determine the transition point time, τ, by minimizing the residual sum of squares from both segments:

$$RSS = \sum_{t=1}^{\tau} (\beta_0 + \beta_1 t + \epsilon_t)^2 + \sum_{t=\tau}^{T} (\gamma_0 + \gamma_1 t + \epsilon_t)^2 \tag{2.31}$$

Relevance to Exchange Rates: Modeling and identifying various trends in exchange rate behavior prior to and following significant economic events or policy changes can be accomplished with piecewise linear regression. This approach allows for a clearer understanding of how exchange rates respond to different economic conditions over time.

2.6.0.5 Bayesian Change Point Analysis

Developed by Isaac and Griffin (1989), Bayesian change point analysis estimates the likelihood that a change point will occur at different positions in the time series by applying Bayesian inference. When the precise location of the transition point is unknown, this approach is especially helpful.

$$P(\theta \mid D) \propto P(D \mid \theta) P(\theta) \tag{2.32}$$

Where:

- $P(\theta \mid D)$: Posterior probability of the change points θ given the data D.
- $P(D \mid \theta)$: Likelihood of the data given the change points θ.
- $P(\theta)$: Prior probability of the change points θ.

When estimating change points, this method enables the integration of past beliefs and uncertainty.

Relevance to Exchange Rates: By taking into consideration the uncertainty around the potential timing of these breaks, Bayesian change point analysis can offer a probabilistic framework for identifying interruptions in exchange rate time series. This approach is particularly helpful when working with noisy or partial data.

2.6.0.6 Role of Change Point Analysis Statistical Tests in Enhancing Exchange Rate Forecasting Accuracy

These statistical tests are essential for identifying and evaluating structural discontinuities in time series of exchange rates, which helps to improve forecasting models. By seeing these shifts, analysts can strengthen their forecasts of the exchange rate and gain insight into the fundamental forces that shape currency movements.

2.7 Model Performance and Model Selection

2.7.1 Importance of Model Performance Evaluation

Assessing the performance of forecasting models is an essential step in guaranteeing their accuracy and dependability, especially when it comes to exchange rate prediction. It is crucial to evaluate how well a model captures trends, patterns, and possible outliers in financial time series data because of its inherent complexity and volatility (CFO Selections Team, 2023). Insights into the model's prediction ability as well as its advantages and disadvantages in different market scenarios are obtained through a thorough evaluation procedure. Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) are examples of performance measures that provide a thorough understanding of a model's forecasting capability, which is essential for making well-informed investments.These metrics are crucial for measuring the accuracy of the model and making sure that predictions match actual facts. According to Hong (2023), performance evaluation informs decision-making processes and offers vital information about how effectively the model is working and whether it achieves the intended goals.

According to Znaczko(2013) effective model evaluation in exchange rate forecasting increases trust in the model's usefulness and relevance. Choosing the appropriate performance criteria is essential for evaluating the model's resilience in a variety of scenarios because exchange rates are impacted by a number of variables, including market sentiment, governmental regulations, and worldwide economic conditions. Additionally, assessing the model's performance across both short- and long-term timeframes can help determine how well it can adjust to abrupt changes in currency dynamics. This stage is essential for choosing the best forecasting model for practical uses, particularly for controlling economic risks and refining currency management plans. Researchers and policymakers may guarantee more accurate forecasts through careful performance evaluation, which will ultimately result in more informed currency market decisions.

2.7.2 Performance Metrics for Forecasting Models

A number of performance metrics must be used in order to assess exchange rate forecasting algorithms. These metrics are useful for evaluating the models' correctness as well as for comparing the applicability of various modeling techniques. Typical performance measurements include the following:

2.7.2.1 Mean Absolute Error (MAE)

Jiang et al. (2017) state that the average of the absolute discrepancies between the observed and projected values is known as the Mean Absolute Error, or MAE. It is a crucial indicator for assessing a model's

total forecast error (Deshpande, 2019). The MAE provides a clear evaluation of the model's accuracy in exchange rate forecasting by calculating the average absolute error between the predicted and actual exchange rates.The following is its formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2.33}$$

where:

- n is the number of observations in the dataset.
- y_i is the true value.
- $\hat{y}_i$ is the predicted value.

2.7.2.2 Root Mean Square Error (RMSE)

According to deepChecks AI (2024), the root mean square error (RMSE) is calculated by taking the square root of the average squared differences between the expected and observed results. RMSE is sensitive to big forecast deviations because it assigns greater weight to larger errors than the Mean Absolute Error (MAE). This feature is especially helpful in situations when significant errors reduce the forecast's usefulness, like in exchange rate forecasting or other volatile markets. The following is the formula for RMSE:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \tag{2.34}$$

where:

- N is the number of data points
- y_i is the i-th true or observed value
- $\hat{y}_i$ is the i-th predicted or estimated value

2.7.2.3 Mean Absolute Percentage Error (MAPE)

According to Roberts (2023), The average magnitude of forecast errors is measured by the Mean Absolute Percentage Error (MAPE), which is expressed as a percentage of the actual values (Roberts, 2023). Because it displays forecast errors in relation to actual values, this metric offers a simple means of evaluating prediction accuracy and facilitates cross-dataset interpretation and comparison. Because MAPE normalizes the error by the actual value, it is extremely helpful when working with time series data, particularly when the scale of numbers varies greatly.

$$\text{MAPE} = \frac{\sum_{i=1}^{N}\left|\frac{A_i - F_i}{A_i}\right| \times 100}{N} \tag{2.35}$$

where:

- N is the number of fitted points
- A_i is the actual value
- F_i is the forecast value
- Σ represents the summation notation, and the absolute value is summed for every forecasted point in time.

2.7.2.4 Akaike Information Criterion (AIC)

The Akaike Information Criterion (AIC) is a statistical technique that penalizes for model complexity while evaluating a model's goodness of fit (Bevans, 2020). By choosing models that strike a compromise between simplicity and fit, the AIC helps prevent overfitting, a major issue in exchange rate forecasting. When a model is overfit, it may perform well on training data but poorly in practical applications. Reliability in exchange rate forecasting depends on the selected model's accuracy and generalizability, which the AIC guarantees by penalizing complexity. This is the AIC formula:

$$\text{AIC} = 2k - 2\ln(\hat{L}) \tag{2.36}$$

where:

- AIC is the Akaike information criterion
- k is the number of estimated parameters in the model
- $\hat{L}$ is the maximum value of the likelihood function for the model

2.7.2.5 Bayesian Information Criterion (BIC)

Analyttica Datalab (2019) states that the Bayesian information criterion (BIC) is a criterion for choosing a model among a limited number of models.Like AIC, the BIC penalizes model complexity more severely. When choosing models with a lot of parameters, it is especially helpful in preventing overfitting.

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L}) \tag{2.37}$$

where:

- BIC is the Bayesian Information Criterion
- n is the number of observations
- k is the number of estimated parameters in the model
- $\hat{L}$ is the maximum value of the likelihood function for the model

2.7.2.6 Diebold-Mariano (DM) Test

It is a statistical hypothesis test that is used to compare the forecast accuracy of two competing models (Tousignant, 2022).When contrasting more sophisticated methods like machine learning models with more conventional models like ARIMA, this test is particularly helpful because it assesses if one model's predicted

performance is noticeably superior to another. When it comes to forecasting exchange rates, the DM test can assist in identifying which model most accurately depicts the underlying dynamics of exchange rates, providing important information for choosing the best forecasting strategy.This is the formula:

$$\text{DM} = \frac{1}{\sqrt{\frac{1}{T}\sum_{t=1}^{T} d_t^2}} \tag{2.38}$$

where:

- $\bar{d} = \frac{1}{T}\sum_{t=1}^{T}(L_1(t) - L_2(t))$ is the average of the loss differential between the two models
- $d_t = L_1(t) - L_2(t)$ is the loss differential at time t, which is the difference between the errors of the two models
- T is the total number of observations (forecast points)

The two models are regarded as equally correct under the null hypothesis, which indicates that the DM statistic has a normal distribution. A DM statistic that is either notably large or little suggests that one model performs better than the other.This formula aids in determining the statistical significance of performance discrepancies between two models.

2.7.2.7 Directional Accuracy (DA)

According to Costantini et al. (2019),Directional Accuracy (DA) is the degree to which a forecasting technique accurately forecasts the direction of a time series. A model's ability to accurately forecast the direction of an exchange rate change—that is, whether it will increase or decrease—is assessed by DA. This measure is especially significant in situations where the direction of change is more significant than its magnitude, like in financial market decision-making processes.

The formula for Directional Accuracy (DA) is:

$$\text{DA} = \frac{1}{T}\sum_{t=1}^{T} I(\text{sign}(\hat{y}_t) = \text{sign}(y_t)) \tag{2.39}$$

where:

- T is the total number of observations (forecast points)
- $\hat{y}_t$ is the predicted value at time t
- y_t is the actual value at time t
- $\text{sign}(\hat{y}_t)$ and $\text{sign}(y_t)$ are the signs of the predicted and actual values, respectively (i.e., whether the value is positive or negative)
- I is the indicator function, which returns 1 if the signs match and 0 if they do not

2.7.3 Factors Influencing Model Selection

Fan (2020) states that a number of factors affect the choice of the best exchange rate forecasting model, including:

1. **Accuracy:** The model's ability to accurately forecast changes in exchange rates is the most crucial factor to take into account. Models with strong predictive power and low mistakes (such as low MAE or RMSE) are preferred. Accuracy by itself, however, is insufficient because highly accurate models may exhibit overfitting or be difficult to interpret.

2. **Complexity:** Predictive accuracy and model complexity are frequently traded off. While simpler models like ARIMA might produce accurate predictions, they might miss intricate, nonlinear dynamics in the data. Although more intricate models, like those used in machine learning, might provide greater accuracy, interpretability and computing efficiency may suffer as a result.

3. **Data Availability:** A major factor in model performance is the availability of high-quality, high-frequency data. While machine learning algorithms can frequently manage irregular, noisy data, models such as GARCH or ARIMA require stationary time series data with predictable trends. Due to its recent inception, the ZiG may have little historical data, which makes simpler models more appropriate.

4. **Scalability and Processing Efficiency:** Computationally efficient and scalable models are preferred when dealing with huge datasets or limited processing resources. Complex machine learning models, which can demand a lot of processing power, are computationally more demanding than traditional time series models like ARIMA.

5. **Interpretability:** Model interpretability is essential in settings that are governed by policies, such as Zimbabwe. Decision-makers must comprehend the reasoning behind a model's predictions. While deep learning models are frequently viewed as "black boxes," traditional models such as ARIMA or GARCH provide greater clarity on their parameters and underlying assumptions.

2.7.4 Challenges in Model Performance Evaluation

The accuracy and dependability of predictions may be impacted by a number of significant issues that arise while evaluating exchange rate forecasting algorithms.

1. **Overfitting:** When a forecasting model becomes too complex and absorbs the noise in the training data, it can perform poorly when applied to unknown data. This is a typical problem in forecasting models (Hastie et al., 2009). When working with small or erratic datasets, like the early phases of a new currency like the Zimbabwean Gold (ZiG), overfitting can be especially problematic in exchange rate forecasting. When external shocks have a major impact on the exchange rate, overfitting may also happen because models find it difficult to adjust to these abrupt changes (Boyd, 2012).

2. **Data Limitations:** The availability of trustworthy, high-quality data is a crucial constraint in exchange rate forecasting. The data may be scarce, non-stationary, or frequently impacted by policy changes, particularly for recently launched currencies like the ZiG. The creation of precise forecasting models may be hampered by this lack of thorough data since it may fail to recognize long-term patterns or take abrupt fluctuations in the exchange rate into consideration (Hong, 2023).

3. **Volatility Clustering:** Exchange rates commonly show this phenomenon, with calmer intervals following times of extreme volatility. Because of this occurrence, modeling becomes more difficult and calls for specific methods like GARCH models, which can account for time-varying volatility. These trends might not be taken into consideration by standard time series models, which could result in estimates that are off during periods of extreme market stress.

4. **External Shocks:** Unexpected external shocks like political unpredictability, abrupt changes in governmental policies, inflationary pressures, or worldwide financial crises frequently have an impact on exchange rate fluctuations (Bordo et al., 2007). These shocks can result in sudden fluctuations in exchange rates that conventional forecasting models might not be able to account for, and they are challenging to model. These shocks' unpredictability draws attention to the shortcomings of forecasting models that do not take them into consideration.

2.8 Empirical Literature

2.8.1 Exchange Rate Forecasting in Other Countries Using Time Series

Due to their capacity to identify trends and temporal relationships in financial data, time series models have been extensively used for forecasting exchange rates in a number of nations. Classical techniques like Generalized Autoregressive Conditional Heteroskedasticity (GARCH) and Autoregressive Integrated Moving Average (ARIMA) have offered fundamental techniques for examining changes in exchange rates. But new methods for time series forecasting have been brought about by machine learning, which makes it possible to predict intricate, nonlinear patterns in financial data more adaptably.The use of both conventional and machine learning-based time series techniques in projecting exchange rates across various nations is reviewed in this section. The methodological framework for examining the ZiG exchange rate will be informed by the findings of these studies, which will provide comparative viewpoints on forecasting techniques used in other nations.

Akincilar et al. (2011) created a forecast model for the Ghanaian cedi's exchange rate against the US dollar using time series analysis. They determined that ARIMA (1,1,1) was the best-fitting model using exchange rate data spanning 1994 to 2010. They created a forecast plot for 2011 and 2012 using this model. Their results showed that the ARIMA model was reliable for short-term forecasting of Ghanaian exchange rates, since the anticipated exchange rates agreed with the series' declining tendency.

Using data from 2013 to 2015, Tran et al. (2016) created an ARIMA model to predict the foreign exchange rate VND/USD (Vietanam currency against the USD currency). The exchange rate between VND and USD for 2016 was predicted using the model. The findings demonstrated that the ARIMA model was appropriate for short-term foreign exchange rate estimation in Vietnam.

The forecast performance of the Autoregressive Fractional Integral Moving Average (ARFIMA) and ARMA models was evaluated by Olanrewaju et al. (2008) when applied to the foreign exchange rate between the US and the UK. They showed that the ARFIMA model outperformed the ARMA model. According to their forecasting evaluation tool, the ARFIMA model is more realistic and accurately depicts the two countries' current economic realities.

Kanyiri (2019) used a Seasonal Autoregressive Integrated Moving Average (SARIMA) model to anticipate foreign currency rates in Kenya, with a particular focus on the USD/KES exchange rate. The study underlined the significance of precise exchange rate projections for assessing foreign-denominated cash flows in international transactions after the implementation of a floating exchange rate regime in 1993. The study determined the main elements driving exchange rate variations, such as supply and demand dynamics, inflation rates, and interest rates, by examining secondary monthly data from January 1993 to March 2019. Based on the Bayesian Information Criterion (BIC) and Akaike's Information Criterion (AIC), the results showed that the SARIMA(1, 1, 0)(0, 0, 2)[12] model had superior performance over the ARIMA(1, 1, 0) model.For short-term forecasting, SARIMA showed the lowest Mean Absolute Error (MAE) of 0.1651, demonstrating its usefulness as a tool for market participants and policymakers negotiating the intricacies of Kenya's foreign exchange market.

In their study, Akhtar et al. (2022) examined the daily exchange rate between the US dollar (USD) and the Pakistani rupee (PKR) using data from January 2018 to April 2021. They first used the ARIMA (1,1,1) model, which was chosen based on performance metrics such as the Akaike Information Criterion (AIC) and Schwarz Criterion (SIC). Although the ARIMA model was able to capture the short-term behavior of the exchange rate series, the presence of volatility clustering required the use of ARCH and GARCH models to account for conditional heteroscedasticity. The results showed that ARCH modeling produced better forecasts than GARCH, as evidenced by lower Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) values.This study emphasizes how crucial it is to combine volatility models with conventional time series models in order to increase the precision of exchange rate forecasts in emerging nations such as Pakistan.

Asad Ullah et al. (2020) used monthly data from January 2011 to December 2020 to analyze the Indian Rupee to US Dollar exchange rate. The nonlinear NARDL model and univariate models (ARIMA, Exponential Smoothing, and Naïve) were part of their strategy. The best univariate model among them was found to be ARIMA (4,1,3), whereas NARDL performed better than all the others with the lowest Mean Absolute Percentage Error (MAPE) of 0.6653. In contrast to individual models, the study showed that combining models—particularly univariate and multivariate ones—often increases predicting accuracy. The benefits of combining linear and nonlinear models for reliable exchange rate forecasts were illustrated by this combination of forecasting technique.

Over a 16-week period from January 3, 2011, to April 22, 2011, Maria and Eva (2011) looked at the daily exchange rates of the Romanian Leu (RON) against six major currencies: the US dollar, British pound, Japanese yen, Chinese renminbi, Russian ruble, and the euro. For short-term forecasting, they used ARIMA models in conjunction with a variety of exponential smoothing approaches, including Holt-Winters methods and Single and Double Exponential Smoothing. Their findings showed that because exponential smoothing approaches can swiftly adjust to fluctuations in the market, they sometimes performed better than ARIMA models. ARIMA models, on the other hand, performed better when anticipating medium-term trends. For instance, ARIMA (1,0,0) produced strong forecasts for EUR/RON and USD/RON, but because of their short-term volatility, exponential smoothing techniques performed exceptionally well for JPY/RON and CNY/RON.This study highlighted how ARIMA and exponential smoothing models work in tandem to estimate exchange rate changes under various market scenarios.

Using data from January 2009 to December 2018, Alshawadfi et al. (2023) projected the Somali Shilling's (SOS) daily exchange rate versus the US dollar (USD). They contrasted Box-Jenkins models with models from the ARCH family, such as TGARCH, EGARCH, and GARCH. According to the Akaike Information Criterion (AIC) and the Schwarz Information Criterion (SIC), the ARMA (0,6) model was determined to be the best Box-Jenkins model. Lower RMSE, MAE, and MAPE values, however, showed that ARCH (6) under the t-distribution was the best-performing model among volatility models because of the presence of volatility clustering and heteroscedasticity. The Diebold-Mariano (1995) test, which highlights that both approaches are useful for many forecasting situations, interestingly showed no discernible difference in the predictive power of the ARMA and ARCH family models.

Tlegenova (2015) used the ARIMA model to predict the Kazakh tenge (KZT) exchange rates from 2006 to 2014 in relation to the US dollar (USD), euro (EUR), and Singapore dollar (SGD). The study emphasized the value of time series analysis in comprehending KZT depreciation patterns, especially during times when the National Bank of Kazakhstan changed its monetary policy. With performance indicators including Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE) proving its accuracy in predicting, the study found ARIMA to be a strong model for capturing the rising trend in exchange rates (Tlegenova, 2023).Similar to findings in other research where ARIMA's efficacy was evaluated against diverse currencies and economic conditions, this work emphasizes the significance of ARIMA in short-term exchange rate predictions and its relevance in emerging countries.

The examined papers demonstrate how well time series techniques capture exchange rate changes, with models such as GARCH and ARIMA providing critical insights. However, the necessity for more sophisticated methods is highlighted by their shortcomings in dealing with nonlinearity and structural changes. Based on these results, established methods can be modified for the ZiG exchange rate, guaranteeing that models are customized for Zimbabwe's economic environment for better forecasting and currency management.

2.8.1.1 Machine Learning-Based Time Series Forecasting of Exchange Rates

Modern machine learning techniques have drastically changed the field of exchange rate forecasting, sometimes outperforming and sometimes supplementing more conventional time series approaches. It has been shown that methods like Artificial Neural Networks (ANNs), Long Short-Term Memory (LSTM) networks, and hybrid models that combine machine learning and statistics perform better at identifying intricate and nonlinear connections in exchange rate data. This subsection reviews the research on the use of time series techniques based on machine learning in exchange rate forecasting, highlighting how these techniques can be more accurate and flexible in a range of economic situations.

Rahman et al. (2021) used daily data from 2015 to 2020 to investigate the application of machine learning algorithms to anticipate the Euro's (EUR) exchange rate versus the US Dollar (USD). Neural network topologies, such as feedforward neural networks (FNN) and long short-term memory (LSTM) models, were contrasted with conventional ARIMA models by the researchers. A baseline for short-term variations was given by ARIMA, but LSTM performed noticeably better, especially when it came to identifying nonlinear patterns. This demonstrates how hybrid modeling techniques, which combine statistical and machine learning techniques to improve exchange rate prediction accuracy, are becoming more and more relevant.

In a thorough investigation of currency exchange rate predictions, Mohammadi (2019) used a number of machine learning methods, such as Support Vector Regression (SVR), Radial Basis Function (RBF), Backpropagation, and LSTM. The study used historical FOREX market data, concentrating on currency pairs such as USD/JPY and EUR/USD. According to the study, SVR performed better than other models, demonstrating how well machine learning captures market dynamics. According to Rahman et al. (2021), this supports the idea that models like as LSTM and SVR are essential for enhancing forecast accuracy in erratic currency exchange markets.

In their 2020 study, Dautel et al. compare Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) with conventional models to investigate deep learning methods for foreign exchange (FX) rate forecasting. Although deep learning has made strides in a number of domains, the authors point out that its use in FX forecasting is still relatively unexplored. Their results underline the significance of model selection in attaining optimal accuracy in the unpredictable FX market by indicating that simpler models might occasionally produce equivalent or better trading profits.

Nguyen et al. (2023) used data from January 2016 to December 2021 to examine the daily exchange rate between the British pound (GBP) and the Japanese yen (JPY). Several forecasting techniques, such as recurrent neural networks (RNNs), SVM, and ARIMA, were compared. Although ARIMA was successful in capturing linear trends, RNNs demonstrated notable gains in forecast accuracy, especially when the market was volatile. This study highlights how machine learning approaches can improve exchange rate predictions by taking into account time-dependent and nonlinear dynamics.

By comparing the Canadian dollar, Australian dollar, British pound, and euro to the US dollar, Tsuji (2022) highlights the potential of the random forest algorithm for exchange rate forecasting. Tsuji shows that the random forest method produces good predicting accuracy by analyzing daily data from December 2009 to June 2022. Because of the limited use of random forest approach in finance, this study promotes more research into machine learning techniques to increase prediction accuracy in dynamic financial situations.

Erem's (2023) study looks at the CHF/EUR, GBP/EUR, and USD/EUR pairs in order to estimate currency rates using artificial neural networks (ANNs). By using a single hidden layer perceptron structure, the study demonstrates that ANNs outperform conventional models like ARIMA and achieve notable accuracy, particularly in daily forecasts. For forex traders and financial authorities, the ability of ANNs to capture complicated exchange rate trends is essential due to their non-linear capabilities. Erem's research emphasizes how machine learning is applicable to financial forecasting, highlighting the value of sophisticated algorithms in ever-changing markets.

Saraswathi et al. (2023) suggested a machine learning algorithm-based approach for forecasting currency exchange rates. With the use of datasets for each pair, the study concentrated on three currency pairs: INR/USD, EUR/USD, and CAD/USD. A hybrid machine learning algorithm that combined five different machine learning models—Multiple Linear Regression, Random Forest Regression, Decision Tree Regression, Gradient Boosting Regressor, and a Voting Regressor—was used. The Hybrid Algorithm outperformed the others in terms of predicted accuracy, obtaining a 99% R2 score. The researchers emphasized the importance of precise forecasts in overseeing global trade and financial stability, but they also pointed out the possibility

of additional improvements through user-friendly platforms and real-time updates. The study emphasizes how hybrid methods might improve exchange rate forecasts.

Küçük (2023) examined a number of machine learning methods for predicting foreign exchange rates, with a particular emphasis on the Turkish Lira, Euro, and Russian Ruble. A variety of models were used in the study to capture both linear and non-linear dynamics in exchange rate behavior, including ARIMA, Extreme Gradient Boosting, Support Vector Regression, Long Short-Term Memory (LSTM), and Convolutional Neural Networks. Notably, the study presented a hybrid model that included ARIMA with LSTM, which demonstrated promise for improving forecasting accuracy. Using measures like Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE), the efficacy of these models was assessed.The results showed that classic benchmarks like the Random Walk model remained competitive, especially for the Euro, even though no single model consistently beat others. The intricacy of forex forecasting and the need for a broad toolkit to adjust to the ever-changing global financial markets were highlighted by this.

In their 2020 study, Ruofan Liao et al. investigated how well machine learning models—more especially, Recurrent Neural Networks (RNN) and Support Vector Machines (SVM)—predict the US dollar's (USD) exchange rate against the Chinese renminbi (RMB). In order to provide a baseline for comparison, the study also used monthly data from 2006 to 2019. In terms of Mean Square Error (MSE), the results showed that RNN performed better than SVM and ARIMA models, especially when optimized using the Bayesian Regularization (BR) technique and set up with six hidden neurons and four layers. This demonstrates RNN's greater capacity to identify dynamic and nonlinear patterns in exchange rate fluctuations.In order to further improve model accuracy and highlight the possibility of combining cutting-edge machine learning techniques with emerging data sources for financial forecasting, the study additionally added the Baidu Index as a fresh input variable. This variable reflects public interest through search engine data.

By tackling the nonlinear and dynamic character of exchange rates, machine learning approaches have proven their capacity to overcome the shortcomings of classic time series models. In Zimbabwe's unstable economic environment, applying these cutting-edge techniques to the ZiG exchange rate can improve forecasting accuracy and aid in initiatives to increase currency stability.

2.8.1.2 Comparison of Traditional and Machine Learning Time Series Models

In financial data analysis, the comparison of contemporary machine learning techniques with conventional statistical methods for time series forecasting has grown in importance. According to Talib et al. (2024), conventional models such as ARIMA are useful for simulating stationary data and linear patterns, but they frequently fail to capture the intricacies and volatility present in financial time series. The capacity of these models to represent rapid market volatility and non-linear interactions is restricted. However, machine learning models like LSTM and GRU are excellent at identifying long-term temporal patterns and non-linear connections, which makes them ideal for dynamic datasets.The paper also emphasizes GRU's computational effectiveness, making it a desirable choice for real-time forecasting in erratic markets where precision and speed are crucial.

In a similar vein, Bakır et al. (2023) examine exchange rate forecasting in unstable economies and note

that while conventional models such as ARIMA function well in times of stability, they falter when volatility is high. On the other hand, during times of market volatility, machine learning models like LSTM and XGBoost can adjust to non-linear behaviors and outside shocks, producing more precise forecasts. The study also highlights the significance of hybrid models, which combine the adaptability of machine learning approaches with the linear trend modeling capabilities of ARIMA. Hybrid models are a crucial tool for predicting exchange rates because of their ability to account for both seasonal trends and sudden market fluctuations, which enables more accurate forecasts.

Zhang and Hamori (2020) contribute to this discussion by showing that forecast accuracy is much increased by combining basic economic theories (e.g., UIP, PPP, and the Taylor rule) with machine learning models like random forest, SVM, and neural networks. Their results highlight how machine learning may capture intricate relationships between economic variables and model non-linearities, which traditional models frequently miss. A strong framework for raising the precision and dependability of exchange rate forecasting is provided by this fusion of theory-driven models and data-driven machine learning approaches. The usefulness of hybrid models in capturing the entire range of market dynamics is highlighted by the durability of their findings, which are backed by assessment measures like Diebold-Mariano tests and RMSE.

By comparing the effectiveness of ARIMA and LSTM on datasets with different seasonal structures, Han (2018) adds even more to the discussion. According to the study, ARIMA performs very well in datasets where linear patterns predominate and there is significant seasonality. LSTM, on the other hand, is better at identifying complex, long-term dependencies and non-linear dynamics in financial data. The study also notes that, particularly when used on simpler datasets, LSTM does not always perform better than ARIMA. This implies that the intricacy of the data and the accessibility of pertinent features have a significant impact on how well machine learning models work. These results demonstrate how well traditional and machine learning techniques complement one another, highlighting the benefits of hybrid models that integrate the two techniques to produce predictions that are more reliable and accurate.

In summary, conventional models such as ARIMA are good at identifying seasonality and linear trends, but they have trouble handling the complexity and volatility of financial data. When it comes to managing non-linear dynamics and market shocks, machine learning models like LSTM and XGBoost provide exceptional flexibility. A more reliable forecasting framework is offered by hybrid models that incorporate both strategies, capturing both complicated market behaviors and linear trends. Exchange rate projections are more reliable when traditional and machine learning techniques are combined, which makes them more appropriate for volatile and dynamic markets like the ZiG/USD exchange rate.

2.8.2 Volatility Modeling of Exchange Rates

A vital tool for comprehending and predicting exchange rate swings, volatility modeling provides information that is important to investors and regulators alike. In order to capture the intricate dynamics of exchange rate volatility, a number of studies have used Autoregressive Conditional Heteroscedasticity (ARCH) and Generalized Autoregressive Conditional Heteroscedasticity (GARCH) models. These studies have emphasized the significance of taking non-Gaussian distributions, leverage effects, and volatility clustering into account.

Epaphra (2017) used the GARCH and EGARCH models to examine the Tanzanian Shilling's (TZS) volatility in relation to the US dollar (USD) between January 2009 and July 2015. His research demonstrated that historical exchange rate movements had a significant impact on volatility, with the data exhibiting both leverage effects and volatility clustering. The limits of conventional models that assume constant variance were brought to light by this study, which also emphasized the significance of using sophisticated volatility models to produce exchange rate projections that are more accurate.

Using many GARCH models, such as ARCH(p), GARCH(p,q), and EGARCH(p,q), Dritsaki (2019) examined the volatility of the Euro/US Dollar exchange rate. In particular, the leverage effect was best captured by the ARIMA(0,0,1)-EGARCH(1,1) model with generalized error distribution (GED), according to the study. Additionally, the study showed that static models produced more accurate predictions than dynamic models, confirming the effectiveness of GARCH models for volatility forecasting.

Todri and Biçoku (2024) used GARCH models for volatility forecasting in their study of the EUR/CHF exchange rate volatility in the Albanian market. They underlined the importance of Value at Risk (VaR) in managing exchange rate risks and pointed out that sustaining financial stability, especially in smaller economies that are more susceptible to changes in foreign exchange, depends on precise volatility forecasts.

Bruno Dinga et al. (2023) used GARCH models to examine the Central African CFA Franc's (XAF) volatility from January 2017 to September 2022 against the US Dollar (USD) and the Chinese Yuan (CNY). Their analysis found that the CNY/XAF exchange rate had notable leverage effects and volatility clustering, whereas the USD/XAF pair did not. By using model selection criteria like the Akaike Information Criterion (AIC), the authors were able to identify the best models for predicting exchange rate volatility in these markets. These models include ARMA(0,1) + GJR-GARCH(1,1) - SGED1.

Omari et al. (2017) used a variety of GARCH models, including symmetric GARCH (1,1), GARCH-M, and asymmetric models including EGARCH (1,1), GJR-GARCH (1,1), and APARCH (1,1), to study the USD/Kenya Shilling (KES) exchange rate from January 2003 to December 2015. The study's findings, which emphasize the significance of taking into account asymmetry in exchange rate volatility, showed that the asymmetric APARCH model, GJR-GARCH, and EGARCH models with a Student's t-distribution offered the greatest fit for the data.

Horvath and Yurttagüler (2023) used effective exchange rate data from 2003 to 2022 using the ARCH and GARCH models to predict exchange rate volatility in Turkey. By addressing the drawbacks of conventional models that assume constant variance and highlighting the importance of time-varying volatility in exchange rate analysis, their findings showed that the GARCH(1,1) model was best suited for simulating exchange rate volatility.

Abdullah et al. (2017) used daily exchange rate data from January 2008 to April 2015 to investigate the volatility of the Bangladeshi taka (BDT) in relation to the US dollar. According to the study, the GARCH(1,1) model offered the best fit to the data, especially when combined with a Student's t-distribution for error terms. The study promoted the use of sophisticated modeling tools to capture the fat-tailed character of financial returns, highlighting the shortcomings of conventional models that assume normally distributed errors.

Using data from June 2001 to June 2023, Gırgin (2023) examined the volatility of the USD/Turkish Lira (TRY) exchange rate. According to the study, different models were most suited for various sub-periods, but the ARMA(4,3)-EGARCH(1,1) model with a skew t-distribution was the most suitable for the total dataset. This study underscores the shortcomings of conventional models that presume normally distributed mistakes, stressing the significance of non-Gaussian distributions in precisely representing the dynamics of exchange rate volatility.

The USD/Mauritian Rupee exchange rate volatility forecasting was investigated by Ng Cheong Vee et al. (2011) using a GARCH(1,1) model, contrasting two distributional assumptions: the Student's-t distribution and the Generalized Error Distribution (GED). Both models produced reliable projections, according to their analysis, which employed daily exchange rate data from June 2003 to March 2008. The GARCH(1,1)-GED model had a modest advantage in out-of-sample predictions. The study emphasized the significance of fat-tailed distributions for capturing the empirical features of financial time series and their applicability to stable economies and efficient risk management.

Marreh et al. (2014) used an ARMA-GARCH modeling approach to study the volatility of Gambian exchange rates, with a particular focus on the Euro and USD against the Gambian Dalasi (GMD). In order to find the best models, the study examined daily exchange rate data from 2003 to 2013 using the Akaike Information Criterion (AIC). Significant volatility persistence was demonstrated by the data, with the ARMA(1,1)-GARCH(1,1) model fitting the USD/GMD the best and the ARMA(2,1)-GARCH(1,1) model fitting the Euro/GMD the best. This study underlined the importance of good volatility modeling for controlling economic risks in The Gambia and the need to include heavy-tailed distributions in financial modeling.

Salisu (2011) studied the naira's volatility in relation to major currencies under various democratic regimes in Nigeria between 1999 and 2011. Using daily return data split into three four-year periods, the study used a GARCH modeling framework. The results showed that, during the first period, the GARCH(1,1) model performed better, but the TGARCH(1,1) model was the best match for the second and third periods. The study's findings of considerable leverage effects—positive shocks increasing volatility more than negative shocks—highlight the need for customized modeling techniques when predicting Nigerian exchange rate behavior.

In summary, when taken as a whole, these studies demonstrate the value of sophisticated GARCH models for simulating exchange rate volatility, especially in very unstable economies. These research' conclusions are directly applicable to the modeling of ZiG exchange rates, providing information on the potential short-term behavior of Zimbabwe's new currency due to non-Gaussian distributions, volatility clustering, and leverage effects. In order to enable efficient policy interventions and stabilize the currency in the face of economic uncertainties, the study of ZiG exchange rates can benefit from a more realistic portrayal of its dynamics through the adoption of these sophisticated volatility models.

2.8.3 Change Point Analysis in Exchange Rates

Change point analysis has been used in a number of studies to analyze exchange rate series, showing that it is a useful tool for comprehending and predicting currency dynamics, particularly during times of market shocks, policy changes, and economic transitions. The following papers provide important insights into the possible use of change point analysis to the Zimbabwean Gold (ZiG) exchange rate, showcasing significant contributions to the field.

Using a time series regression model with GARCH errors, Hardi et al. (2015) applied change point analysis to the Indonesian Rupiah (IDR) against the USD, focusing on the years 2000 to 2008. The authors found structural changes in the IDR's relationship with the USD, especially around March 2002, reflecting changes in Indonesia's exchange rate management. Their methodology, which combined log-likelihood ratio tests and cumulative sum of squares of residuals with Monte Carlo simulations, successfully detected important exchange rate shifts. The study's methodology could be applied similarly to detect shifts in the ZiG exchange rate, driven by policy changes or external economic factors, providing a valuable tool for exchange rate forecasting.

Cruz et al. (2022) used Bayesian change point analysis to examine Chile's USD/CLP exchange rate between 2018 and 2020. The authors found abrupt changes in the dynamics of the exchange rate by applying Markov Chain Monte Carlo (MCMC) techniques and a semiparametric Bayesian model. Their research showed that the suggested model successfully represented sudden shifts in the exchange rate mean, which is essential for comprehending currency swings. This study is very pertinent to the ZiG exchange rate since Bayesian techniques can be used to detect comparable abrupt changes that might arise in reaction to monetary policy changes and Zimbabwe's changing economic environment.

In their 2021 study, Awiakye-Marfo et al. examined the convergence of a randomized change point estimator in GARCH models using daily USD/GHS exchange rate data from 2008 to 2019. It was shown by their study that the randomized pseudolikelihood ratio change point estimator was capable of detecting important shifts in the variance structure of exchange rate data. GARCH models' forecasting capabilities were improved by the study by offering a strong foundation for change point detection, especially when structural alterations were included. Since identifying changes in volatility could greatly enhance forecasting and risk assessment in light of Zimbabwe's economic difficulties, the findings are extremely pertinent to the examination of the ZiG currency rate.

Using change point analysis, Mungai et al. (2020) investigated the Kenyan Shilling's (KES) volatility in relation to other major currencies. To identify notable changes in the KES exchange rate, the authors used statistical techniques like the likelihood ratio test and the CUSUM test. The results showed significant shifts in the value of the KES, impacted by both internal and external economic policy. This study illustrates how change point analysis may be used to comprehend currency performance in developing nations, which makes it pertinent to the examination of the ZiG exchange rate, which is subject to fluctuations from both internal and external sources.

Dette et al. (2019) addressed the drawbacks of conventional techniques that rely on stationarity by introducing a framework for change point analysis in non-stationary time series. They suggested employing

piecewise locally stationary processes, which enable the identification of variations in the variance, higher-order moments, and time-varying mean. Using this technique, the authors were able to pinpoint second-order change points associated with volatility shifts in the USD/CAD exchange rate series. The ZiG exchange rate is a prime example of an adaptive approach to non-stationary time series, as it may exhibit non-stationary behavior as a result of changes in governmental policy or the economy. By using this technique, forecasting accuracy might be increased and the dynamics of volatility in the ZiG exchange rate better understood.

Using empirical likelihood techniques, Machuka (2023) used sequential change point estimate to examine the KES/USD exchange rate between January 2017 and December 2021. under order to improve forecast accuracy, the study showed that the empirical likelihood approach successfully detected change points under turbulent market conditions. By validating the estimator's asymptotic qualities through Monte Carlo simulations, a reliable tool for real-time financial analysis was made available. This strategy is quite useful for the ZiG exchange rate, particularly when volatility is strong, since it provides a quick way to identify sudden shifts and improve risk management procedures.

These studies demonstrate how vital insights into how abrupt changes in a currency's dynamics might affect its volatility and forecasting accuracy can be gained by using change point analysis to exchange rates. These approaches are especially pertinent to the ZiG exchange rate because of Zimbabwe's history of unstable currency and the likelihood of sudden fluctuations in its value. In order to better capture structural changes and create more precise models for predicting exchange rate fluctuations, this research will include change point analysis into the ZiG study. The end result of this strategy will be better currency risk management plans for Zimbabwe.

2.9 Conclusion

This chapter examined the main theoretical and empirical studies that are pertinent to predicting and forecasting the ZiG/USD exchange rate. Time series modeling, volatility modeling, and change point analysis were among the fundamental ideas and theories that were covered first, then empirical research on exchange rate forecasting in different nations. The chapter stressed the value of traditional time series methods, machine learning techniques, and hybrid approaches in a variety of economic scenarios. It was noted that volatility modeling and change point analysis are crucial for identifying distinctive characteristics of the ZiG exchange rate, namely structural breaks and volatility clustering.Finally, a conceptual framework that was customized to the particular difficulties of the ZiG currency was developed thanks to insights into model evaluation metrics and selection criteria. These results set the stage for the following chapter, which will build on this review with the study design and analytical techniques.

Chapter 3

Methodology

3.1 Introduction

This chapter provides a comprehensive discussion of the research methodologies employed to achieve the research objectives outlined in Chapter 1. Through the use of time series approaches, volatility modeling, and change point analysis, these methodologies aim to predict and forecast the Zimbabwe Gold (ZiG) exchange rate. Exploratory data analysis (EDA), data preparation procedures, and the particular modeling methodologies used are explained after the study strategy is described at the beginning of the chapter. Additionally covered are software tools, ethical issues, and diagnostic procedures to guarantee the validity, dependability, and robustness of the findings. This chapter primarily aims to defend the selected methodologies and show how applicable and successful they are at capturing the short-term fluctuations of the ZiG/USD exchange rate.

3.1.1 Research Strategy

A quantitative approach is used in the study, which is ideal for the empirical examination of numerical data. Through the use of statistical and economic methods, this approach makes it possible to identify trends, patterns, and structural shifts in the ZiG/USD exchange rate. The study aims to analyze short-term movements in the ZiG/USD exchange rate, identify structural breaks using change point analysis, and assess the exchange rate's volatility dynamics. These goals drove the choice of this approach. The objective and repeatable nature of quantitative research is crucial for the analysis of time series data in financial environments. A balanced approach to accomplishing the study's goals is provided by the strategy, which combines contemporary machine learning techniques with conventional econometric models.

3.1.2 Study Population

This is the entire collection of information, people, groups, or subjects pertinent to the goals of a study.Daily exchange rate data for the ZiG/USD currency pair, beginning with the currency's launch in April 2024, make up the research population. The dataset contains the daily bid and ask rates in addition to the midrate (ZiG/USD exchange rate). The bid-ask spread, a helpful tool for volatility modeling and spotting changes in market dynamics, is determined by the bid and ask rates, which also offer extra information about market liquidity and volatility.This dataset captures the short-term exchange rate changes and the

ZiG's performance during its early adoption period.

Scope of the Data:

1. **Timeframe:** From April 2024, when the ZiG was introduced, to the present.
2. **Granularity:** Daily exchange rates to effectively capture short-term fluctuations.
3. **Source:** Reserve Bank of Zimbabwe (RBZ) portal, ensuring accuracy, reliability, and credibility of the data.

The study's goals are in line with this dataset's exceptional suitability for examining short-term trends, examining volatility clustering, and spotting structural discontinuities in exchange rate behavior.

3.2 Data Preparation

A crucial first step in making sure the dataset is organized and ready for analysis is data preparation. The following steps are involved in the process:

1. **Data Collection**

    a. I'll access the Reserve Bank of Zimbabwe (RBZ) portal to obtain the ZiG/USD currency pair's daily exchange rate information. The following columns will be part of the dataset:

        i. **Date:** Indicates the precise day when the exchange rate was noted.
        ii. **Bid Rate:** The rate at which the market or bank is willing to buy the ZiG.
        iii. **Ask Rate:** The rate at which the market or bank is willing to sell the ZiG.
        iv. **ZIG/USD Mid Rate:** The average of the bid and ask rates, representing the exchange rate used for analysis.

    As a standardized indicator of the exchange rate, the midrate will be the main focus of forecasting. In order to improve the dataset's relevance, I will also use derived features such as:

    1. **Bid-Ask Spread:** This feature measures market liquidity and is calculated as:

    $$\text{Bid-Ask Spread} = \text{Ask Rate} - \text{Bid Rate} \tag{3.1}$$

    The **Bid-Ask Spread** will be used in **volatility modeling** and **change point analysis** to assess liquidity stress and market dynamics.

    2. **Month:** Extracted from the Date column to identify seasonal patterns. The **Month** feature will be applied in **SARIMA models** and **exploratory data analysis (EDA)** to assess recurring monthly trends.
    3. **Daily Percentage Change:** Captures short-term volatility and is calculated as:

    $$\text{Daily Change (\%)} = \frac{Y_t - Y_{t-1}}{Y_{t-1}} \times 100 \tag{3.2}$$

    In this formula, **Midrate** represents the exchange rate on day t (current day) and $t-1$ (previous day). This feature will be helpful for **volatility modeling** and detecting significant market fluctuations during **change point analysis**.

4. **Rolling Averages:** Highlights short-term trends, calculated as:

$$\text{Rolling Avg (k days)} = \frac{1}{k} \sum_{i=t-k+1}^{t} Y_i \tag{3.3}$$

Here, k represents the window size (e.g., 7 days), and t represents the current day. **Rolling Averages** will smooth out fluctuations and assist in identifying underlying trends, which will be crucial for **forecasting models**.

5. **Volatility Index:** Quantifies market risk using rolling standard deviation:

$$\text{Volatility (k days)} = \sqrt{\frac{1}{k} \sum_{i=t-k+1}^{t} (Y_i - \text{Rolling Avg (k days)})^2} \tag{3.4}$$

In this formula, **Rolling Avg (k days)** is the rolling average over k days, and t represents the current day. The **Volatility Index** will play a key role in **volatility modeling** (e.g., GARCH models) to measure market risk over specific periods.

6. **Lagged Values:** Represents historical dependencies, such as:

$$\text{Lag}_n = Y_{t-n} \tag{3.5}$$

In this formula, **Lag** represents the exchange rate from n days before the current day t. I will use **Lagged Values** as inputs in **time series models** (e.g., ARIMA, SARIMA, LSTM) to capture the relationship between current and past exchange rate values.

An Excel file containing the dataset will be saved for later preparation and analysis.

2. **Data Cleaning:**

- **Analysis of Outliers:**The researcher will employ boxplots and Z-scores to identify outliers—exceptional values that significantly deviate from the overall trend. The origin and significance of the identified outliers will be investigated. They will be dealt with if needed to prevent them from distorting the analysis.

3. **Stationarity Testing:**

- I will use the Augmented Dickey-Fuller (ADF) Test to ascertain whether the data is stationary for the use of time series models. The ADF test's null hypothesis presupposes non-stationarity; its failure would suggest stationarity.

  If the dataset turns out to be non-stationary, I will use first-order differencing to stabilize the mean and variance, using the following formula:

$$\Delta Y_t = Y_t - Y_{t-1} \tag{3.6}$$

  Where:

  - ΔY_t: The differenced value at time t
  - Y_t: The value at time t

- Y_{t-1}: The value at the previous time point

4. **Feature Engineering** To improve model performance, the following features will be developed from the raw data:

    1. **Bid-Ask Spread:** In order to capture market liquidity dynamics, this derived characteristic will be crucial for **volatility modeling** and **change point analysis**.
    2. **Month:** Will support **seasonal trend detection** and **SARIMA modeling**.
    3. **Daily Percentage Change:** This feature is crucial for predicting **volatility** and spotting abrupt changes in the market.
    4. **Rolling Averages:** These enhance **forecasting models'** accuracy by highlighting short-term trends and mitigating volatility.
    5. **Volatility Index:** In volatility modeling, the **volatility index** is used to quantify risk over a range of time periods.
    6. **Lagged Values:** In order to reflect historical dependencies, **lag values** are essential in **time series models**.

5. **Normalization:**

- For machine learning models,I'll normalize the data to make sure all the features are on the same scale in order to get it ready for machine learning models. I'll use min-max scaling, which is described as:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{3.7}$$

  Where:

  - X: The original value
  - $X_{\min}$: The minimum value of a variable
  - $X_{\max}$: The maximum value of a variable
  - X_{scaled}: The scaled value

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a method of understanding the underlying structure and properties of a dataset.

3.3.1 Descriptive Statistics

I will calculate crucial descriptive metrics to comprehend the exchange rate data's fundamental tendencies and variability:

i. **Mean and Median:** To ascertain central tendency

ii. **Standard Deviation and Variance:** To measure the range of changes in the exchange rate.

iii. **Range:** To record the lowest and highest values noted throughout the research period.

iv. **Skewness and Kurtosis:** To evaluate the exchange rate distribution's symmetry and tail behavior.

3.3.2 Visualizations

The use of visual aids will be essential for spotting trends and irregularities. I'll make use of the visualizations listed below:

1. **Time Series Plot:** This helps me identify trends, cycles, and possible outliers by visualizing the **ZiG/USD exchange rate's** daily variations over time.
2. **Histogram:** To display the exchange rate's frequency distribution and assist in determining its shape (normal, skewed, etc.).
3. **Charts of Autocorrelation and Partial Autocorrelation (ACF and PACF):** These charts will help determine the time series' lag dependencies. While **PACF** will enable me to ascertain the order of the time series models (e.g., ARIMA) by displaying direct correlations after adjusting for the intervening lags, **ACF** will assist in evaluating the correlation of the series with its own lagged values.

3.4 Time Series Modelling

Time series modeling enables the study of patterns including trends, seasonality, and cyclical behaviors by evaluating and forecasting data points indexed in time order. It is especially pertinent to exchange rate studies, as daily observations can highlight volatility, short-term swings, and structural changes. The models and techniques used to predict the ZiG/USD exchange rate are described in this section.

3.4.1 Traditional Time Series Models

Traditional time series models work well with univariate datasets, such as the ZiG/USD exchange rate, because they are based on statistical techniques. Since the statistical characteristics (mean, variance, and autocorrelation) are assumed to be constant throughout time, these models rely on stationarity in the data as explained by Petropoulos(2021).

3.4.1.1 ARIMA Model

I will analyze the ZiG/USD exchange rate's historical fluctuations using the ARIMA model to estimate short-term changes. Since the midrate is the standardized reference rate, it will be the main input for the model (ZiG/USD exchange rate). For short-term financial and economic decision-making, ARIMA's ability to simulate these linear correlations will be useful in predicting future exchange rate levels. The model's efficacy is derived from its capacity to identify trends and cyclical fluctuations.The formula for **ARIMA** is:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \epsilon_t - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots \tag{3.8}$$

Where:

- Y_t: Represents the exchange rate value at time t.
- ϕ_1, ϕ_2: Autoregressive coefficients, which indicate the influence of previous values on the current value.
- θ_1, θ_2: Moving average coefficients, which show how previous forecast errors influence the current value.

- ϵ_t: The error term at time t, capturing the random disturbance at each point.

The exchange rate data's linear dependencies are captured by this formula. In order to provide precise short-term projections of the **ZiG/USD exchange rate**, the **ARIMA model** will be essential for predicting the future exchange rate based on these historical associations.

3.4.1.2 Seasonal ARIMA (SARIMA)

Al-Gounmeein (2020) states that the Seasonal ARIMA (SARIMA) model is an extension of the ARIMA model that accounts for seasonal fluctuations in time series data by incorporating additional autoregressive (AR) and moving average (MA) elements as well as seasonal differencing.The formula for **SARIMA** is:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \theta_1 \epsilon_{t-1} + \epsilon_t + \text{Seasonal Components} \tag{3.9}$$

Where the **Seasonal Components** capture the seasonal trends.

If there are predictable seasonal factors, such as changes in government policy or market activity linked to particular months, I will use **SARIMA** to model and anticipate recurrent seasonal trends in the **ZiG/USD exchange rate**.

3.4.1.3 Exponential Smoothing (ETS)

It is a forecasting technique that exponentially decreases the weight of previous observations while increasing the weight of recent data. **ETS** does not require stationary data, in contrast to **ARIMA**. The ETS formula is:

$$S_t = \alpha Y_t + (1 - \alpha) S_{t-1} \tag{3.10}$$

Where:

- S_t: The smoothed value at time t.
- α: The smoothing factor.

I'll use **ETS** to find underlying trends and seasonality while mitigating short-term swings in the **ZiG/USD exchange rate**. **ETS** is perfect for producing accurate short-term forecasts since it works especially well when the data is erratic or volatile.

3.4.2 Time Series Machine Learning Models

Traditional statistical methods can be replaced by machine learning models, particularly when it comes to identifying long-term dependencies and non-linear correlations.

3.4.2.1 Long Short-Term Memory (LSTM) Networks

Long-term dependencies in sequential data can be captured by LSTM networks, a kind of recurrent neural network (RNN). When it comes to displaying complex, non-linear patterns in exchange rate data, they are excellent.

I will estimate short-term changes in the ZiG/USD exchange rate using LSTM networks, taking into account both non-linear patterns and historical trends. Furthermore, I will use elements like the Bid-Ask Spread to represent the dynamics of market liquidity that impact swings in exchange rates.

3.4.2.2 Random Forest and XGBoost

Random Forest: This ensemble method aggregates predictions from multiple decision trees, reducing overfitting and improving robustness.
XGBoost: An advanced gradient boosting technique, efficient for large datasets with complex relationships I will compare the performance of **ARIMA** and **SARIMA** using **Random Forest** and **XGBoost**, paying particular attention to how extra features like the **Bid-Ask Spread** improve model accuracy. These models will make it possible to identify intricate links in the data that conventional approaches could miss.

3.4.2.3 Hybrid Models

Statistical time series techniques like ARIMA and machine learning techniques like LSTM, Random Forest, and XGBoost are combined in hybrid models to increase forecast accuracy. The forecasting performance of hybrid models can be greatly improved by utilizing both machine learning models' capacity to detect non-linear patterns and ARIMA's capacity to detect linear trends.

1. **ARIMA + LSTM Hybrid:** In this hybrid, **LSTM** predicts the non-linear fluctuations while **ARIMA** captures the linear trends. In order to enable the hybrid model to capture both linear and non-linear components of the exchange rate data, the residuals (errors) from ARIMA will be fed into LSTM.By incorporating both linear and non-linear patterns, this hybrid model will help me forecast the **ZiG/USD exchange rate** more accurately.

2. **ARIMA + XGBoost Hybrid:** While **XGBoost** captures non-linear linkages and interactions, such as those between the **Bid-Ask Spread** and lagged exchange rates, **ARIMA** captures the linear trends in this hybrid model.By fusing XGBoost's ability to handle intricate, non-linear interactions with ARIMA's ability to simulate linear trends, I will use this hybrid model to improve the **ZiG/USD exchange rate** forecast.

3. **ARIMA + Random Forest Hybrid:** In order to capture non-linear dependencies and feature interactions, like lagged values and bid-ask spread, this hybrid combines **Random Forest** with **ARIMA** for linear trend modeling.I will utilize this hybrid model to enhance **ZiG/USD exchange rate** predictions by leveraging Random Forest's adaptability in identifying intricate patterns and ARIMA's linear trend modeling.

3.4.3 Steps in Time Series Modeling

1. **Stationarity Testing**
   The researcher will use the Augmented Dickey-Fuller (ADF) Test to determine whether the series is stationary. I will use differencing to stabilize the mean and variance if the series is determined to be non-stationary, making sure the data is appropriate for ARIMA/SARIMA models.

2. **Optimal Lag Selection**
   I will utilize the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC) to find the ideal lag duration for ARIMA models. The model will be better able to identify dependencies in the data if the lag is carefully chosen.

3. **Forecasting**
   Once the model has been fitted, the researcher will produce predictions for future exchange rates. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are two measures I'll use to assess how accurate these projections are.

3.5 Volatility Modeling

An essential tool for assessing the risk and uncertainty included in exchange rate fluctuations is volatility modeling. Volatility clustering is a common phenomenon in exchange rate data, when high volatility periods are followed by comparable high or low volatility periods.This study incorporates both bid-ask spread and historical exchange rate data to better capture market dynamics and liquidity effects in volatility modeling.The researcher will use the following models to illustrate and comprehend these dynamics:

3.5.0.1 GARCH (1,1) Model

The GARCH (1,1) model is frequently applied to financial data that exhibits volatility over time. Since it assumes that present volatility is a function of past volatility and prior error terms, it is particularly helpful in identifying volatility clustering, in which little changes typically follow small ones and large changes typically follow large changes.

For GARCH (1,1), the formula is:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{3.11}$$

Where:

- σ_t^2: Represents the conditional variance (volatility) at time t.
- ω: The constant term (intercept).
- α: The coefficient for past squared errors (ϵ_{t-1}^2).
- β: The coefficient for past volatility (σ_{t-1}^2).
- ϵ_t: The error term, representing the deviation from the forecast at time t.

I'll utilize the **GARCH (1,1)** model to pinpoint times when the **ZiG/USD exchange rate** is more volatile. In order to improve the efficacy of this model, I will include a number of derived features, including rolling averages, volatility index, bid-ask spread, daily percentage change, and lagged values. These attributes will be employed to capture liquidity stress, short-term volatility, and historical dependencies, thereby enhancing our understanding and prediction of market conditions.

3.5.0.2 EGARCH Model

I'll use the EGARCH model, which is an expansion of the fundamental GARCH framework, to identify asymmetries in the ZiG/USD exchange rate's volatility, especially when it comes to negative shocks. This model will provide a more realistic depiction of market behavior during volatile times by integrating factors

such as the bid-ask spread, daily percentage change, and lagged values. These features will also assist capture the influence of unfavorable news on market volatility.The formula for it is as follows:

$$\ln(\sigma_t^2) = \omega + \alpha(\sigma_{t-1}\epsilon_{t-1}) + \beta \ln(\sigma_{t-1}^2) \tag{3.12}$$

Where:

- $\ln(\sigma_t^2)$: The logarithm of the conditional variance (volatility) at time t.
- ϵ_{t-1}: The lagged error term.
- σ_{t-1}^2: The lagged conditional variance (volatility).
- α: Captures the impact of past shocks (scaled by past volatility).
- β: Measures the persistence of volatility from previous periods.

This logarithmic representation allows for a more stable estimation of the volatility dynamics and helps in capturing the relationship between past volatility and current estimates.

3.6 Change Point Analysis

Change point analysis makes it possible to detect structural breaks by identifying periods when a major change takes place in the underlying process affecting the exchange rate. It is essential to comprehend how changes in policy or macroeconomic events will affect the ZiG/USD exchange rate over the long run. This analysis will help identify significant changes that impact the ZiG/USD rate's stability, especially when there are external market impacts, policy changes, or economic shocks.Through the use of elements like bid rate, ask rate, and bid-ask spread, I hope to record changes in market liquidity and volatility dynamics in addition to fluctuations in exchange rate levels. We'll employ the three tests listed below:

3.6.0.1 Chow Test

When there appears to be a structural break at a particular point in the exchange rate series, the **Chow Test** is employed. This test compares regression coefficients before and after the break point to ascertain whether they differ considerably.

The formula for the Chow Test is:

$$F = \frac{(RSS_1 + RSS_2)/(n_1 + n_2 - 2k)}{(RSS_R - (RSS_1 + RSS_2))/k} \tag{3.13}$$

Where:

- F: The test statistic.
- RSS_R: The residual sum of squares for the restricted model (no break).
- RSS_1 and RSS_2: The residual sums for the two separate regression models (before and after the break).
- k: The number of coefficients.

- n_1 and n_2: The number of observations before and after the break.

I will use the **Chow Test** to find noteworthy changes in the **ZiG/USD exchange rate**, especially in reaction to important policy changes like the currency rate announcement in September 2024. We'll also include the **bid-ask spread** to account for shifts in market liquidity.

3.6.0.2 CUSUM Test

Significant changes in a time series' mean over time can be found using the **CUSUM Test**. It is particularly effective for recording unexpected market disruptions or policy changes, as it facilitates the identification of rapid changes.

The formula for the CUSUM Test is:

$$S_t = \sum_{i=1}^{t} (Y_i - \bar{Y}) \tag{3.14}$$

Where:

- S_t: The cumulative sum at time t.
- Y_i: The observed value at time i.
- $\bar{Y}$: The mean of the series.

Using the **CUSUM Test**, the researcher will look for long-term changes in the **ZiG/USD exchange rate**, especially in relation to events like the policy change in September 2024. Incorporating the **bid-ask spread** will aid in capturing the effects of shifts in liquidity during noteworthy occurrences, such as expanding spreads that signify market stress.

3.6.0.3 Bayesian Change Point Detection

Using Bayesian reasoning, the Bayesian Change Point Detection method determines the likelihood of structural changes in the time series. Finding shifts that happen at erratic or unpredictable intervals is made easier with this technique. The development of a probabilistic framework provides insight into the causes and timing of these shifts.

I will analyze ZiG/USD exchange rate swings using this approach, especially when the swings are erratic. In order to comprehend market liquidity dynamics, the Bayesian Change Point Detection will also use elements like the bid-ask spread to help capture changes in market behavior before to or following structural breaks.

3.6.0.4 Bai-Perron Test

Multiple structural breaks at unknown times in a time series can be found using the Bai-Perron Test. This test is flexible for finding several breaks throughout the series because it does not presume a set duration for each break. The Bai-Perron Test formula is as follows:

$$\hat{\beta}_j = (X'X)^{-1}X'Y \tag{3.15}$$

Where:

- $\hat{\beta}_j$: Represents the estimated regression coefficients for the break at j.
- X: The matrix of independent variables, which includes the midrate, bid rate, ask rate, bid-ask spread, and any lagged values that capture historical dependencies.
- Y: The dependent variable, which is the **ZiG/USD exchange rate**.

I will utilize the **Bai-Perron Test** to identify several structural breaks in the **ZiG/USD exchange rate**, especially in relation to events such as policy or macroeconomic developments. The test will include the bid-ask spread, bid rate, ask rate, and lagged data to help evaluate how these factors affect exchange rate dynamics over time.

3.7 Model Diagnostic

Model diagnostics play a critical role in validating the performance and reliability of forecasting models for the ZiG/USD exchange rate. They ensure that the models adequately capture the underlying patterns and meet key assumptions such as independence, stationarity, and normality of residuals. Diagnostic tests such as the Ljung-Box Q-Test are employed to assess autocorrelation in residuals, ensuring no significant patterns are left unexplained(hassani,2020). The Durbin-Watson Statistic evaluates serial correlation, confirming the independence of residuals, while the ARCH-LM Test detects conditional heteroscedasticity, a key feature in volatility modeling. These tests collectively enhance model robustness, ensuring that forecasts are accurate and reliable for decision-making in financial contexts.

3.8 Software Used

A variety of software tools are used in this study to make data processing, analysis, and documentation easier. Python is the main tool used for machine learning model implementation, exploratory data analysis (EDA), and data preprocessing. Its libraries, including scikit-learn, pandas, matplotlib, and TensorFlow, provide effective data visualization, manipulation, and the creation of sophisticated models like Long Short-Term Memory (LSTM) networks. Utilizing packages like fGarch and forecast to create GARCH models and run statistical tests like the Augmented Dickey-Fuller (ADF) and Ljung-Box Q-Test, R is used for statistical modeling and volatility analysis.Excel is also utilized for the initial arrangement and storing of data, offering a simple platform for recording the daily exchange rate data from the Reserve Bank of Zimbabwe (RBZ) portal. Additionally, LaTeX will be utilized to typeset the dissertation, guaranteeing expert formatting and smooth incorporation of tables, figures, and mathematical formulas. This set of tools guarantees a thorough and effective workflow, utilizing the advantages of each platform to meet the goals of the study.

3.9 Ethical Considerations

The study complies with strict ethical standards to guarantee honesty and openness throughout the investigation. The only source of the exchange rate information is the Reserve Bank of Zimbabwe (RBZ) portal, which is a reliable and open source. For the sake of academic rigor and reproducibility, all data, techniques, and algorithms used are properly attributed. The focus of the study is on objective analysis that does not involve data modification or fabrication. The validity of the results is maintained by handling anomalies

and outliers in a methodical manner. Confidentiality is maintained despite the data being publicly available by appropriately presenting the findings without raising unnecessary concerns or damaging stakeholders' reputations. The credibility of the research is strengthened by these ethical procedures, which are in line with professional and academic norms.

3.10 Conclusion

This chapter offered a thorough approach that covered the research plan, study population, data preparation, and analytical procedures for evaluating and predicting the ZiG/USD exchange rate. It presented machine learning and time series models for forecasting, emphasized the value of change point analysis and volatility modeling, and explained diagnostic procedures to confirm the models' dependability. The study's software tools were also described, and ethical issues were covered to guarantee accountability and openness. These methodological procedures support the validity and rigor of the study findings and provide a strong basis for the analysis that follows.